

GENERAL THETA FOUNDATIONS I: ACQUIRED GEOMETRY AND CAUSAL RESOURCE TRANSFER

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ABSTRACT. A prepared causal experiment determines predictive information through executable future tests, but a finite observer can retain only laws generated by its own acquisitions and one executable programme. We give an exact allocation criterion for a single parameter-independent programme on a family of compact predictive experiments. Its affine dual couples all occupied parameters and phases before selecting a next label. On jointly dominated finite-dimensional positive instruments, a robust optimum is attained and every strict global risk obstruction is already present in finitely many models. With uniformly effective support and continuity certificates, the same finite-label optimum admits computably convergent upper and lower bounds. The construction preserves legal actions, exact stopping and programme reuse; it does not require contraction of normalized filters or interchange the programme infimum with a parameter supremum. Untruncated Gaussian models with nonmixing transitions and noncommuting quantum instruments with unknown channels verify the same theorem. Singular acquired laws, atomic tails and rank changes are included. A calibration experiment has an exact minimax memory threshold. Matching geometric resource curves and common-task simulation are retained with separate accounts for labels, programme description, workspace, precision, acquisitions and time. The effective search is finite but need not be efficient.

1. INTRODUCTION

A causal experiment supplies legal preparations, reports, interventions and executable future tests, not a pre-existing parameter manifold. Histories are predictively equivalent when those tests have identical continuation laws and interfaces. After this quotient has been realized, finite observation remains a causal problem: the law available for compression is generated by the observer's own previous acquisitions, while its retained information determines the next acquisition. Neither the dimension of a formal state space nor a terminal codebook settles whether the required information can be retained and used by one finite programme.

There is a further constraint when the physical model is inaccessible. A programme synthesized separately at each possible model is not a programme that can operate without knowing that model. The same action row, report row, stopping rule and readout must work at every occupied parameter and phase. A conditional predictive state may vary with both of these indices as an analyst's certificate, but neither index is automatically a machine register. In particular a parameterwise Bayes action is not a common minimax readout. The mathematical objective is an exact resource–risk criterion with these quantifiers in their operational order, followed by a finite construction that approximates the criterion at the original memory budget.

Theorem 3.1 is the central result. For a jointly dominated family of prepared compact predictive experiments, it eliminates the common report-to-label kernel through the simultaneous affine inequality (5). The maximum over next labels comes after both the parameter and phase sums.

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This gives an exact acquired-law invariant

$$R_{\text{aut}}^{\Theta}(n, M) = \Phi^{\Theta}(n, M) = \inf_{\text{one feasible common flow}} \sup_{\theta \in \Theta} R_{\theta}.$$

The invariant retains actual conditional occupancies and predictive barycenters; it is not a formula in ambient dimension. The observer does not receive the true parameter, its posterior profile, or an uncharged clock. The single-law barycentric theory is proved in the later sections as a constituent of this statement, not used to authorize a parameter-dependent programme.

On a common finite-dimensional positive physical realization, Theorem 3.3 strengthens this exact criterion: the common-programme optimum is attained, and a strict target below it is excluded by finitely many parameter models. This global statement follows from compactness of the shared report rows through their integrated instrument moments. Reuse of a row creates finite products of these moments, not a discontinuous multiplication of arbitrary pointwise weak-star limits. The finite-model obstruction concerns every programme at the budget, rather than one proposed flow. Without further effective data it supplies neither the number of models nor an algorithm to find them.

The effective clause closes that distinction under explicit uniform presentation hypotheses. A common report reconstruction, support-resolving partition and parameter modulus give finite numbers L, U satisfying

$$L - E_h - \beta_Q \leq \Phi^{\Theta}(n, M) \leq U + \Omega_n(s), \quad U - L \leq \eta.$$

An explicitly enumerated dyadic programme attains the upper bound at the same M persistent labels. The report error E_h , programme/readout resolution β_Q , parameter-net error $\Omega_n(s)$ and integration tolerance η are distinct and computably vanish. This bounds the optimum of *all* Borel programmes at that budget, not merely the best element of a promising finite list. A constant unknown model is used throughout each evaluation. Neither a fresh adversarial model at each call nor the exchange $\inf \sup = \sup \inf$ is invoked. The number of programmes and the offline parameter-net evaluations are explicit; the result is not an efficient planning theorem.

The proof operates on positive unnormalized instruments. Their joint physical-state/label laws have a direct-sum base norm contracted by every common stochastic programme. Averaging a report row against a parameter- and phase-independent reconstruction therefore preserves a common row and gives a uniform error without differentiating a normalized posterior. Support certificates retain the possible legal and stopping paths; dyadic rounding can remove an edge but cannot create a forbidden one. The same norm comparison is uniform over all discontinuous Borel report rows and supplies the parameter modulus needed for global lower as well as upper certificates. This separates predictive-update instability from finite-horizon programme approximation.

Theorems 4.1 and 4.2 verify the same theorem directly for two different families of raw experiments. Gaussian observations are untruncated and their transitions may be reducible or deterministic. The quantum family permits noncommuting, unknown channels and pure preparations, while the observer remains a classical finite programme rather than a quantum memory. Corollary 4.3 treats singular reference laws, controlled atomic tails and flagged changes of rank with their actual masses. The inherited geometry theorems additionally give matching regular and singular resource curves when their sharper mass, covering, curvature or blind-transport assumptions hold. None of those realization conclusions is a premise of the general transfer theorem.

Proposition 4.4 supplies a sharp finite test of the new quantifier. A noisy calibration bit is followed by a fair report whose interpretation depends on the fixed unknown bit. The optimal common programme has an exact five-state threshold, whereas a programme furnished with that bit can have zero risk with four states. The lower bound covers randomized updates and the calibration-offset term is attained in the same task. This example distinguishes information

patterns; it is not offered as a difficult filtering problem solved by elementary algebra. The earlier stationary-reuse parallelogram and noncontracting acquired-law bottleneck remain separate sharpness witnesses.

Comparison of experiments, convex order, conditional martingale kernels and finite-state-controller synthesis are established subjects[3, 17, 24, 23]. Robust synthesis for uncertain POMDPs also predates this work[28]; hidden-model controller extraction and verification provide a recent computational comparison[29]. The claim here is not priority for robust finite memory, conditional barycenters, weak-star compactness or finite enumeration. It is the exact shared-parameter/shared-phase elimination, the global finite-model consequence in the positive-instrument category, and its uniformly certified same-budget implementation with continuous reports and a fixed physical parameter. Section 14 separates these assertions from neighboring finite-controller optimization, model learning, discounted control and zero-delay coding results.

The general passage remains

$$\begin{aligned} \text{prepared experiments} &\longrightarrow \text{executable predictive quotients} \longrightarrow \text{causal allocations} \\ &\longrightarrow \text{acquired geometry} \longrightarrow \text{finite-resource risk}. \end{aligned}$$

The exact compact category and the effective positive-instrument category are stated separately. The latter requires a known effective family, not knowledge of its true member. No universal data-driven identification rate, arbitrary noncompact quotient theorem, infinite-horizon limit, or jointly optimal programme/workspace/time region is asserted. Programme bits, numerical approximation, physical acquisitions and parameter-independent simulator state are separately charged. These boundaries leave the mother problem intact while specifying the part for which complete transfer and implementation proofs are given.

2. EXPERIMENTAL DATA AND OBSERVERS

We use $\text{TV}(P, Q) = \sup_B |P(B) - Q(B)| = \frac{1}{2} \|P - Q\|_1$. Fix a finite horizon $n \geq 1$. A call takes a hidden physical state x , a legal action a , and produces a new hidden state, a report, and a resource increment through a positive normalized kernel. Positive means nonnegative and normalized; deterministic reports and zero likelihoods are permitted. Preparation is fixed. An unknown parameter, when present, is not an argument available to the observer. This section first defines a single prepared law. Adjoining an unknown parameter to the hidden state gives a Bayes experiment, not automatically a frequentist minimax theorem. Section 3 supplies the additional common-programme conditions and quantifiers for the latter.

An executable future test is a legal continuation with a bounded scored outcome. At a fixed phase, histories are equivalent if all such tests have the same conditional expectation and the same legal interface. Appendix A gives sufficient raw-kernel hypotheses for a compact metrizable realization of this quotient. In particular it establishes measurability, actual mixture realizability, and minimality. We now state precisely the properties used by the main theorem.

For each $t = 0, \dots, n$, let K_t be a compact metrizable convex subset of a Hausdorff locally convex space, embedded by a countable separating family of continuous affine future-test expectations. Write $\text{Aff}(K_t)$ for continuous affine functions, including constants. Probability measures on K_t have barycenters in K_t , denoted $\bar{\nu}$, and $f(\bar{\nu}) = \int f d\nu$ for $f \in \text{Aff}(K_t)$. No finite dimension is assumed. Let $\mathcal{A}$ be finite and let $\mathcal{A}_t \subseteq \mathcal{A}$ be a fixed nonempty legal action set at phase $t < n$. Let $\mathcal{Y}_a$ be standard Borel. These sets are part of the physical interface, not a phase variable readable by a stationary machine. Kernels for illegal actions may be extended arbitrarily and never enter a positive-mass term. For $t < n$, the raw instruments induce jointly Borel maps

$$(1) \quad J_t(q, a)(dy), \quad F_t(q, a, y) \in K_{t+1}, \quad \Lambda_t(q, a) = \text{Law}_{Y \sim J_t(q, a)} F_t(q, a, Y).$$

Values of F_t on zero-likelihood reports are irrelevant and are assigned measurably. The instrument property means that, for every probability ν on K_t , Borel report set B , and $f \in \text{Aff}(K_{t+1})$,

$$(2) \quad J_t(\bar{\nu}, a)(B) = \int J_t(q, a)(B) \nu(dq),$$

$$\int_B f(F_t(\bar{\nu}, a, y)) J_t(\bar{\nu}, a)(dy) = \int \int_B f(F_t(q, a, y)) J_t(q, a)(dy) \nu(dq).$$

The identities are required on all mixture states in the carrier. They express physical mixing before an unnormalized instrument, not linearity of the normalized Bayes update. Assume $q \mapsto \Lambda_t(q, a)$ is weakly continuous. This ensures continuous Bellman values; the feasibility parts below only need the displayed measurable instruments and conditional barycenters. Total-variation report continuity is an additional quantitative property verified in the regular realizations, not a premise of convex order.

The terminal decisions form a compact metric space U . The scored loss has conditional expectation $\ell(q, u) \in [0, H]$, jointly continuous and affine in q for each u . Let

$$(3) \quad g(q) = \min_{u \in U} \ell(q, u), \quad V_n = g, \quad V_t(q) = \min_{a \in \mathcal{A}_t} \int V_{t+1}(w) \Lambda_t(q, a)(dw), \quad B_n^* = V_0(q_0).$$

Compactness, weak continuity and finite actions give continuous values and Borel minimizing decisions/actions. Concavity follows by taking infima of the affine risks of fixed continuation programmes: a programme may depend on its subsequently observed reports, but is fixed before the incoming preparation is mixed. Its affine risk need not be continuous for this argument. The instrument identities make that mixture comparison valid. The terminal task and hence B_n^* may change with the horizon.

Definition 2.1 (Two finite-state interfaces). An externally clocked observer has finite sets Z_t , with $|Z_0| = 1$ and $|Z_t| \leq m_t$ for $1 \leq t \leq n$. It uses action probabilities $\alpha_t(a \mid z)$ and a Borel stochastic kernel $k_t(z' \mid z, a, y)$. The external phase t is free in this model. Terminal decisions depend only on Z_n .

An autonomous observer is a *single* tuple $(Z, z_0, D, \alpha, k, u)$: $|Z| \leq M$, $D \subset Z$ is the set of stopping labels, $\alpha(a \mid z)$ is one time-independent action kernel on $Z \setminus D$, $k(z' \mid z, a, y)$ is one time-independent next-label kernel, and $u : D \rightarrow U$ is one terminal readout map. At a stopping label it issues the readout and makes no further call. Starting from z_0 , it must stop after exactly n calls almost surely. Neither kernel may consult absolute time unless it is encoded in z . Fresh independent randomization is erased at the next persistent cut. A retained pseudorandom-generator seed, instruction pointer, output tape or clock is part of Z .

A reused autonomous label has one action distribution supported in the intersection of all phase-legal action sets at which it has positive occupancy. The common report space is indexed by action, not by an unread phase. Programme cardinality here bounds labels only; a general Borel rule need not have a finite-bit description. Theorem 8.2 supplies one effective specialization.

A fixed read-only programme can depend on the known preparation, kernels, horizon and budgets, but not on the acquired data or an inaccessible parameter. Programme length, temporary workspace, precision, physical calls and physical time are separate resources. Exact-real measurable programmes define the cardinality theorem; finite-bit implementation is addressed in Section 13. An unused label may have arbitrary programmed behavior. Legality means $\alpha_t(\mathcal{A}_t \mid z) = 1$ on every positive-mass phase label; for a stationary observer the same α_z must obey this requirement at every phase at which z occurs. All measures below are generated by the actual deployed observer, including its fresh randomization.

Write $R_{\text{ext}}(\mathbf{m})$ and $R_{\text{aut}}(n, M)$ for the infimal risks in these two classes, with infimum $+\infty$ for an empty class. Write $R_{\text{ext}}(n, M)$ when every $m_t = M$. The checkpoint relaxation permits

full history during acquisition and restricts only the final label to M values. These are different information patterns, all using the same preparation and task.

Lemma 2.2 (Affine witnesses and executable tests). *On a compact predictive carrier K , the uniform closure of the linear span of the constant and the determining affine test coordinates is $\text{Aff}(K)$. A finite expression $c_0 + \sum_{i=1}^m c_i f_i$ can be realized as the expectation of a signed bounded score by randomly selecting one of the executable tests f_i , provided they share a legal continuation budget. The score bound and test-selection cost must be included. A uniform limit is not thereby one finite-cost experiment, and a maximum of affine expectations is not generally an executable single-test expectation.*

In the local allocation inequality, uniform approximation of every affine component within ϵ changes the difference of its two sides by at most 2ϵ . In the simultaneous inequality this bound is $2n\epsilon$. Thus every strict finite-coordinate dual violation is preserved by sufficiently accurate finite test-span approximation, without identifying a convex dual function with one physical continuation.

Proof. Let W be the uniform closure of the stated span. If a continuous affine f were not in W , Hahn–Banach and the representation of continuous linear functionals on $C(K)$ would give a finite signed measure μ annihilating W but not f . Since constants lie in W , its positive and negative parts have equal mass. If that mass is positive, their normalized probabilities have equal determining coordinates and therefore the same barycenter. Affinity gives equal integrals of f , a contradiction. Zero mass is immediate. Uniform limits of affine functions are affine, proving equality.

For a finite expression put $C = \sum_i |c_i|$. If $C > 0$, choose i with probability $|c_i|/C$, execute its test, and score $c_0 + C \text{sgn}(c_i)T_i$, where T_i is the actual test outcome with conditional expectation f_i . For normalized $|f_i| \leq 1$ the score has absolute value at most $|c_0| + C$. The required physical budget is the common allowed budget, with its selector and programme costs, not an assertion that arbitrary collections of differently typed tests have one cheap execution. If $C = 0$ only the constant is needed.

The maximum of finitely many functions is 1-Lipschitz in their uniform component norm. The two local measures have mass one. In the simultaneous expression, the left total occupancy mass is n , while $\sum_{z,a} \eta_{z,a}(\mathcal{V}_a) = n$ and the Radon–Nikodym phase weights sum to one. Applying the same bound on each side gives $2n\epsilon$. $\square$

3. ONE CAUSAL PROGRAMME FOR A PREPARED FAMILY

The quantifier on the programme is essential when an experimental parameter is inaccessible. A separate feasible allocation at each parameter need not be implementable by one observer. We first give an exact common-programme criterion and then an effective, two-sided value theorem. The parameter is fixed before preparation and remains fixed throughout all calls. In particular, taking a worst case anew at each call is a different experiment.

3.1. The common parameter and phase constraint. Let Θ be a nonempty parameter set. At each θ , assume the prepared compact predictive experiment of Section 2, with carrier $K_{\theta,t}$, preparation $q_{\theta,0}$, instrument $(J_{\theta,t}, F_{\theta,t})$, and terminal loss ℓ_θ . The finite action sets, action-indexed report spaces, horizon n and decision space U are common. The modeller knows this family; the observer does not receive θ . Assume, in addition, that for each action there is one finite Borel measure σ_a dominating every $J_{\theta,t}(q, a)$ at a legal phase and state. This joint domination is an additional hypothesis, not a consequence of parameterwise standard-Borel realization. The effective classes below have it. Normalizing a nonzero σ_a is harmless.

Write $\mathcal{G}_{\text{aut}}^\Theta(n, M)$ for the tuples of Definition 2.1 which are legal and stop at n under every θ . All their components, including their terminal readouts, are parameter independent. Define

$$(4) \quad R_{\text{aut}}^\Theta(n, M) = \inf_{\gamma \in \mathcal{G}_{\text{aut}}^\Theta(n, M)} \sup_{\theta \in \Theta} R_\theta(\gamma).$$

An empty class has value $+\infty$. The analogous external class allows phase-indexed rows, but not parameter-indexed rows. Terminal readouts are deterministic in the declared space U . If terminal lotteries are allowed, they are included in U , with their expected loss and sampling cost; no purification across parameters is assumed.

Fix Z, z_0, D , one action row α_z , and one readout u_j for each $j \in D$. A proposed family of occupancies consists of $p_{\theta,t,z}$ and $q_{\theta,t,z} \in K_{\theta,t}$ satisfying the initial, probability, exact-stopping and legality constraints (25) for every θ . Thus a used action at z is legal at *every* occupied parameter–phase pair. Set

$$h_{\theta,t,z,a} = \frac{d[p_{\theta,t,z}\alpha_z(a)J_{\theta,t}(q_{\theta,t,z}, a)]}{d\sigma_a}.$$

These are nonnegative integrable functions. For a finite set $\mathcal{I} \subset \Theta \times \{0, \dots, n-1\}$ and $\phi_{\theta,t,j} \in \text{Aff}(K_{\theta,t+1})$, consider

$$(5) \quad \begin{aligned} & \sum_{(\theta,t) \in \mathcal{I}} \sum_j p_{\theta,t+1,j} \phi_{\theta,t,j}(q_{\theta,t+1,j}) \\ & \leq \sum_{z \notin D, a} \int \max_j \left\{ \sum_{(\theta,t) \in \mathcal{I}} h_{\theta,t,z,a}(y) \phi_{\theta,t,j}(F_{\theta,t}(q_{\theta,t,z}, a, y)) \right\} \sigma_a(dy). \end{aligned}$$

The maximum is outside *both* the parameter sum and the phase sum. For external widths, use fixed sets Z_t , common rows $\alpha_{t,z}$ and k_t , and require at each fixed t , for every finite $F \subset \Theta$,

$$(6) \quad \sum_{\theta \in F, j} p_{\theta,t+1,j} \phi_{\theta,j}(q_{\theta,t+1,j}) \leq \sum_{z,a} \int \max_j \left\{ \sum_{\theta \in F} h_{\theta,t,z,a}(y) \phi_{\theta,j}(F_{\theta,t}(q_{\theta,t,z}, a, y)) \right\} \sigma_a(dy).$$

In this external display h uses $\alpha_{t,z}$ in place of α_z , and the same terminal readout is used across parameters. Let $\mathfrak{F}_{n,M}^\Theta$ denote the autonomous proposals satisfying (5) and the stated common-component constraints. Define the actual-law invariant

$$(7) \quad \Phi^\Theta(n, M) = \inf_{\mathfrak{F}_{n,M}^\Theta} \sup_{\theta} \sum_{j \in D} p_{\theta,n,j} \ell_\theta(q_{\theta,n,j}, u_j).$$

This is a risk-valued invariant. It does not separately minimize an oracle baseline or the terminal compression term. In particular, u_j cannot be replaced by a different Bayes action at each parameter. The conditional predictive states are certificate variables under each actual law, not information furnished to the observer.

Theorem 3.1 (Parameter-uniform causal acquired-geometry transfer). *For the dominated prepared family above, a proposed occupancy family is realized by one parameter-independent autonomous observer if and only if it belongs to $\mathfrak{F}_{n,M}^\Theta$. Any failed proposal has a strict witness (5) involving finitely many parameter–phase pairs and affine tests. Consequently*

$$(8) \quad R_{\text{aut}}^\Theta(n, M) = \Phi^\Theta(n, M).$$

For external widths the same statement holds with a common row at every fixed phase; the maximum then remains outside the parameter sum at that phase. No compactness of the parameter set is required for this exact assertion. Attainment is not asserted on arbitrary compact predictive

carriers. If the family instead has a common finite-dimensional positive-instrument realization with joint domination and continuous compact terminal readouts, the optimum is attained and

$$(9) \quad R_{\text{aut}}^{\Theta}(n, M) = \sup_{\emptyset \neq F \subset \Theta, |F| < \infty} R_{\text{aut}}^F(n, M).$$

An empty common class or a strict suboptimal risk target is then excluded already by finitely many fixed models. This conclusion needs neither continuity in the parameter nor normalized-filter stability. Theorem 3.3 specifies and proves this positive-instrument clause.

Under the additional uniformly effective presentation and parameter moduli of Definition 3.2, this invariant has the following finite, globally valid certificates at the same persistent budget. Choose a report partition, dyadic denominator Q , terminal loss net, a finite s -net $\Theta_s \subset \Theta$, and evaluation tolerance $\eta > 0$. Enumerate the admissible programme set $\mathcal{G}_{h,Q,\rho}$ of Theorem 8.2, using its common support graph. For each γ and $\theta_i \in \Theta_s$, obtain

$$l_{\gamma,i} \leq R_{\theta_i}(\gamma) \leq u_{\gamma,i}, \quad u_{\gamma,i} - l_{\gamma,i} \leq \eta, \quad L = \min_{\gamma} \max_i l_{\gamma,i}, \quad U = \min_{\gamma} \max_i u_{\gamma,i}.$$

If the class is nonempty, then

$$(10) \quad \boxed{L - E_h - \beta_Q \leq \Phi^{\Theta}(n, M) \leq U + \Omega_n(s), \quad U - L \leq \eta.}$$

A programme minimizing $\max_i u_{\gamma,i}$ works on the original family without learning θ for free. Its worst-case excess over the optimal M -label Borel programme is at most

$$(11) \quad E_h + \beta_Q + \Omega_n(s) + \eta.$$

The list has at most $M2^M|U_{\rho}|^M(Q+1)^P$ members, with P as in (42); each needs at most $|\Theta_s|$ certified risk evaluations. These certificates converge effectively from both sides. They do not merely bound the best programme in a selected finite list. Empty-class detection is exact under the support hypotheses. The corresponding external assertion uses (45) and its error budget.

The geometric content of (8) is the simultaneous realization of *actually acquired* conditional laws with one finite programme. The quantitative content of (10) is a modulus-controlled converse to finite implementation. Neither replaces the acquired law by the ambient parameter dimension. The report net controls approximation of the physical interface, whereas the parameter net certifies a worst case over inaccessible models. Their sizes are computational costs, not substitutes for the risk geometry. The singleton-parameter theory and its matching regular and singular estimates are proved in Sections 5–12.

3.2. Proof of the exact common-programme statement. For fixed proposed occupancies, let $\mathcal{K}$ be the product, over $z \notin D$ and a , of the simplex-valued unit balls

$$\{(k_j)_j \in L^{\infty}(\sigma_a)^Z : k_j \geq 0, \sum_j k_j = 1\}.$$

It is convex and weak-star compact: positivity and the row sum are closed inequalities and equalities against $L^1(\sigma_a)$. Map $\mathcal{K}$ to the product of all parameter–phase mass and affine-moment coordinates by

$$(12) \quad k \mapsto \left(\sum_{z \notin D, a} \int k(j \mid z, a, y) h_{\theta,t,z,a}(y) f(F_{\theta,t}(q_{\theta,t,z}, a, y)) \sigma_a(dy) \right)_{\theta,t,j,f}.$$

Each coordinate is weak-star continuous, since its coefficient is integrable. The image is compact and convex in the product topology, including when Θ is uncountable. If the target coordinates $p_{\theta,t+1,j} f(q_{\theta,t+1,j})$ are not in this image, a finite-coordinate projection already separates them. Indeed an open basic product neighbourhood of the target misses the compact image; in its finite projection the target is outside a compact convex set. Finite-dimensional separation then gives a

linear combination of those coordinates. Absorb the finitely many scalar coefficients into affine functions $\phi_{\theta,t,j}$, using zero functions for omitted labels and including constants for masses.

The support function of (12) is exactly the right side of (5): at each (z, a, y) choose a maximizing next label *after* adding all the coefficients. The least maximizing label is a measurable choice, so the integral of the maximum is attained. Thus all inequalities hold precisely when a common $k \in \mathcal{K}$ has every prescribed coordinate. On the standard Borel report spaces choose Borel representatives, completing the simplex on one σ_a -null set per row. Joint domination makes this completion harmless for every θ , not only for a chosen countable subfamily.

Now start at $q_{\theta,0}$, for any fixed θ . Assuming the proposed occupancy and conditional predictive state at cut t , apply the fixed action row and the common k . The mass coordinates in (12) give $p_{\theta,t+1,j}$. The remaining coordinates and the unnormalized instrument identity give every future-test expectation at the conditional barycenter $q_{\theta,t+1,j}$. Separating affine tests identify that barycenter. Induction realizes the whole proposal under its own deployed law. The zero-mass restrictions enforce the exact deadline and all positive-probability action restrictions. No parameter-indexed selection of a programme has occurred. Conversely, the conditional barycenters of a genuine programme obey these moment identities, and its rows are bounded by the same support function. Finally the shared terminal readouts have exactly the risks in (7), proving (8). The external proof keeps a separate row for each permitted phase and repeats this separation across parameters.

For example, consider one constant report under two models $\theta = 0, 1$, and propose terminal label $j = \theta$ with certainty. Each separate model admits that label allocation. The constant tests $\phi_{\theta,j} = \mathbf{1}_{\{j=\theta\}}$ make the joint inequality read $2 \leq 1$, exposing the missing common programme. Maximizing separately over labels inside the parameter sum would incorrectly accept it.

A witness is a mathematical comparison of predictions at hypotheticalal models. It is not an experiment permitted to query the true parameter. Lemma 2.2 gives the relation of each affine component to executable test spans. Uniform test-span approximations within ϵ change the two-sided difference for a finite index set $\mathcal{I}$ by at most $2|\mathcal{I}|\epsilon$, since each parameter-phase occupancy has mass one. The finite witness for a *proposal* is distinct from the finite global risk exclusion obtained below.

3.3. Uniform physical presentation and proof of effectiveness.

Definition 3.2 (Uniform effective presentation). The set Θ is a computably compact metric space with supplied finite s -nets of uniformly computable points in Θ . All experiments have a common finite-dimensional ordered physical space (V, V_+, e) with compact normalized base $\mathcal{B}$, as in Definition 8.1. Preparations $x_{\theta,0}$, positive instruments $\mathbf{l}_{\theta,t}^a$, and losses $L_{\theta,u}$ have $0 \leq L_{\theta,u} \leq He$. Computation of their cell integrals and losses is uniform in θ . The cell partitions, reconstruction measures, support tables and terminal loss nets of Definition 8.1 are common to the entire family. In particular, an active cell has equivalent report null sets for every normalized physical state and every parameter, and an inactive cell has zero instrument. A supplied $d_{t,h}$ bounds (41) uniformly in θ .

For $d_{\Theta}(\theta, \theta') \leq s$, supplied computable moduli tending to zero give

$$(13) \quad \begin{aligned} \|x_{\theta,0} - x_{\theta',0}\|_{\mathcal{B}} &\leq \omega_0(s), \\ \sup_{a,x \in \mathcal{B}} \int \|(\mathbf{l}_{\theta,t}^a - \mathbf{l}_{\theta',t}^a)(dy)x\|_{\mathcal{B}} &\leq \omega_t^I(s), \\ \sup_{u,x \in \mathcal{B}} |L_{\theta,u}(x) - L_{\theta',u}(x)| &\leq \omega^L(s). \end{aligned}$$

Define

$$(14) \quad \Omega_n(s) = \min \left\{ H, H \left(\omega_0(s) + \sum_{t < n} \omega_t^I(s) \right) + \omega^L(s) \right\}.$$

The errors E_h, β_Q are those of (42), with a terminal loss net uniform in both state and parameter. Cell access, probability sampling and readout computation have the separately declared execution costs. Merely knowing that a family is continuous does not supply these certificates.

Theorem 3.3 (Attainment and global finite-model obstructions). *Suppose the raw family has a common finite-dimensional ordered physical space with compact normalized base, finite action alphabet, standard Borel reports jointly dominated by finite measures σ_a , and positive normalized instruments. Let the common readout space U be compact metric, with each $u \mapsto L_{\theta,u}$ continuous on the physical space. The parameter set can be arbitrary. Every nonempty common M -label, exact- n class has a worst-case-risk minimizer, possibly requiring a noncomputable Borel programme. Equation (9) holds. If the common class is empty, a finite subfamily already has empty class. If $r < R_{\text{aut}}^\Theta(n, M)$, there is a finite subfamily for which every admissible M -label programme has risk greater than r at some one of its models. The same statements hold for fixed external widths. No bound on the size of this finite subfamily follows without additional data.*

Proof. Pad every programme to M labels. There are finitely many choices of its initial label and stopping subset. Fix one choice and take the product of its action simplexes, its simplex-valued $L^\infty(\sigma_a)$ report rows, and U^M . This is compact in the product of Euclidean, weak-star and readout topologies. A finite disjoint union over the discrete choices remains compact.

For phases at which an action is illegal, use a zero instrument only to extend the ambient occupancy formula; legality constraints exclude every execution of that extension. Thus all coefficient measures used below are dominated by the specified reference measures. Choose a basis of the finite-dimensional physical space. Every coefficient measure of $\mathfrak{l}_{\theta,t}^a$ has an $L^1(\sigma_a)$ density. To see this, decompose a basis vector as a difference of positive vectors. Positivity bounds the total variation of its image by the sum of their masses. On a σ_a -null report set every positive image has zero mass and hence is zero, by strict positivity of the base functional; linearity handles the basis vectors. It follows that each matrix entry of

$$A_{\theta,t,z,a,j}(k) = \int k(j \mid z, a, y) \mathfrak{l}_{\theta,t}^a(dy)$$

is a weak-star continuous function of the common row. For a fixed θ , propagation of the subnormalized physical-state/label vectors obeys

$$v_{\theta,t+1,j} = \sum_{z \notin D,a} \alpha_z(a) A_{\theta,t,z,a,j}(k) v_{\theta,t,z}.$$

It is a finite composition of continuous matrix and scalar operations, even when the same report row is reused. Stopping labels may be made absorbing with an arbitrary positive identity continuation solely to define these continuous functions on the entire ambient programme space. On exactly timed programmes this extension has no effect before the last cut.

At each parameter, absence of a premature stop, absence of an illegal action, and total stopping mass one at the deadline are closed equalities in these continuous occupancy functions. Intersect them over all parameters. This gives a compact admissible set. On it $R_\theta(\gamma) = \sum_j L_{\theta,u_j}(v_{\theta,n,j})$ is continuous for each fixed θ . Its supremum over parameters is lower semicontinuous and thus attains its minimum on every nonempty compact admissible set. Nothing here requires the parameter to be known to the programme.

For a finite target r , let $C_\theta(r)$ be the closed subset of the ambient compact programme space satisfying the legality and deadline constraints at θ , and $R_\theta(\gamma) \leq r$. If every finite intersection of

these sets is nonempty, compactness gives a point in their entire intersection. Contrapositively, a strict target below the robust optimum is excluded by a finite set of models. The same argument without the risk constraint proves finite exclusion of an empty class. Conversely the common optimum is at least every finite-subfamily optimum. Applying the just-proved finite-intersection implication to any number strictly above the supremum of those optima proves (9), including the empty-class case. The external case has finitely many separate phase rows and the same argument. This is a global exclusion of all common programmes, not only separation of one proposed occupancy flow. It does not interchange the programme infimum and parameter supremum. $\square$

Lemma 3.4 (A uniform modulus over all programmes). *For every common admissible Borel programme γ , at every fixed persistent budget,*

$$(15) \quad |R_\theta(\gamma) - R_{\theta'}(\gamma)| \leq \Omega_n(s) \quad (d_\Theta(\theta, \theta') \leq s).$$

The constant is independent of the programme, including discontinuous report rows. It does not assume contraction of normalized predictive updates.

Proof. The physical-state/label law at a cut is a direct sum of positive subnormalized vectors (v_z) . Its norm is $\sum_z \|v_z\|_{\mathcal{B}}$. A positive mass-preserving instrument, followed by any stochastic action and report rows, contracts this norm on signed inputs: apply it to a positive decomposition and sum the preserved masses. At a fixed positive normalized input, replacing the instrument changes the next direct sum by at most $\omega_t^I(s)$, since the same report row is used and its nonnegative entries sum to one. Starting with the preparation difference, a telescoping replacement over all n calls bounds the final norm by $\omega_0(s) + \sum_t \omega_t^I(s)$. The common readout has loss bounded by H in dual norm; changing its physical loss functional adds at most $\omega^L(s)$. Both risks lie in $[0, H]$, proving the clipped bound. One parameter is used throughout each comparison; no timewise adversary or hidden-state trajectory for a quantum instrument is postulated. $\square$

Proof of the effective part of Theorem 3.1. For any common Borel row $k(j \mid z, a, y)$, average it against the *same* reconstruction measure on each report cell. The resulting $\bar{k}(j \mid z, a, b)$ is independent of both phase and parameter. The positive direct-sum comparison in Lemma 8.3, with its error uniform in θ , changes every parameter's risk by at most E_h . The support argument there now applies to every parameter simultaneously; therefore the one cell programme is legal and exactly timed throughout the family. Round each common row once on its original support and choose uniform-net readouts. Lemma 8.4 adds at most β_Q to every risk and cannot create a forbidden path. Thus, if $R_G^\Theta = \min_{\gamma \in \mathcal{G}_{h,Q,\rho}} \sup_\theta R_\theta(\gamma)$, then

$$R_{\text{aut}}^\Theta \leq R_G^\Theta \leq R_{\text{aut}}^\Theta + E_h + \beta_Q.$$

This comparison holds for arbitrarily close Borel competitors and does not need an attained Borel optimum. The common support table proves exact emptiness equivalence with the finite list, for every supplied partition and $Q \geq 1$.

Put $S_G = \min_\gamma \max_{\theta_i \in \Theta_s} R_{\theta_i}(\gamma)$. The net is a subset of the family, and Lemma 3.4 is uniform in γ , so

$$S_G \leq R_G^\Theta \leq S_G + \Omega_n(s), \quad L \leq S_G \leq U.$$

For a programme γ minimizing $\max_i l_{\gamma,i}$, every upper endpoint is at most the corresponding lower endpoint plus η . Hence $U \leq L + \eta$. The risk of a minimizer of $\max_i u_{\gamma,i}$ is at most $U + \Omega_n(s)$ on the whole family. Combining the inequalities gives (10) and (11). All individual risks are computed from the finite compositions of cell instruments, with a single θ_i in each entire composition. The candidate count is unchanged from Theorem 8.2; the parameter net multiplies only the number of evaluations. Letting the four independently certified errors tend to zero proves effective convergence. At no point is $\inf_\gamma \sup_\theta$ interchanged with $\sup_\theta \inf_\gamma$. $\square$

Corollary 3.5 (Finite exclusion, misspecification and causal simulation). *For a strict target $r < R_{\text{aut}}^\Theta(n, M)$, sufficiently fine certificates satisfy $L - E_h - \beta_Q > r$ and thus exclude every admissible M -label Borel programme. The witness consists of a complete finite programme list, parameter net, risk enclosures and the uniform comparison bounds; it is not one affine continuation test.*

If the true fixed parameter lies within δ of the modelled set in an ambient family satisfying the same support conditions and moduli, the attained upper bound increases by at most $\Omega_n(\delta)$. A report-implementation mismatch of total probability at most χ , uniformly along the ideal deployed laws and preserving the support signature, adds at most $H \min(1, \chi)$; terminal numerical error adds its uniform loss modulus.

Suppose one parameter-independent causal simulator of a target family uses at most S persistent states and $C(n)$ source acquisitions, respects the exact action/stopping/readout interface, and approximates the complete stopped and scored law within total variation ε , uniformly over parameters and legal target observers. Then

$$(16) \quad R_{\text{source}, \leq C(n)}^\Theta(MS) \leq R_{\text{target}}^\Theta(n, M) + H\varepsilon,$$

where the source uses at most $C(n)$ calls and the source stopping interface is exactly the one in the simulator certificate. An exact- $C(n)$ source class may replace it only if that exact completion, including its clock and any padding calls, is certified. A reverse lower comparison requires a reverse simulator, not merely a one-way defect bound.

Proof. Convergence of both endpoints gives the strict exclusion. The modulus lemma compares a fixed output programme at a true parameter and a nearby modelled one. The first-discrepancy argument bounds numerical mismatch on the ideal prefix law; signature preservation is required for exact admissibility, rather than inferred from a small error probability. The simulator executes one target tuple with the product of its states and its own states. Bounded-loss total-variation comparison gives $R_\theta^{\text{source}} \leq R_\theta^{\text{target}} + H\varepsilon$ for every θ . Take the supremum and then infima over arbitrarily close target competitors. These are upper comparisons; allowed error tolerances are not automatically attained statistical lower bounds. $\square$

The new minimax statement does not assert an unknown-kernel learning rate. It optimizes a parameter-independent observer over a specified, uniformly presented family, allowing whatever adaptive acquisition and learning its counted states can perform. An oracle full-history baseline depending on the true parameter cannot be subtracted as an attainable robust baseline. An excess-risk claim requires a separately specified and certified robust full-history benchmark for the same family and task. The interval theorem is for total risk and does not need such a subtraction. The singleton benchmark argument in Corollary 8.5 is retained at its stated scope; it is not permission to subtract the parameterwise oracle optimum from an attained robust risk.

For the table representation, put $b = \log_2 Q$ and $K_\rho = |U_\rho|$. The common programme needs at most

$$P(b+1) + M \lceil \log_2 K_\rho \rceil + M + \lceil \log_2 M \rceil$$

bits for the probability table, readout indices, stopping flags and initial label, in addition to the declared cell and readout descriptions and fixed interpreter. The parameter net and candidate enumeration are offline costs, not retained data. Linear row sampling needs $O(b + \log(MA \max_a B_a))$ temporary bits, in addition to cell/readout scratch, and $O((A+M)(b+1))$ elementary bit operations per call in the stated read-only random-access table convention. Address formation and random access are included in that convention; a sequential-tape interpreter must instead pay its actual lookup time. Every location or random seed surviving a persistent cut is included in Z . These are feasible implementation bounds, not lower bounds on optimal programme size, workspace or planning time.

4. RAW VERIFICATION FOR INACCESSIBLE PARAMETER FAMILIES

The following verifications use the unnormalized physical instruments. In particular, an unstable normalized posterior does not invalidate the comparison of all fixed programmes. The physical model is supplied for offline synthesis; its actual parameter and evolving state are not observer registers.

Theorem 4.1 (Unknown, possibly nonmixing Gaussian models). *Let Θ be computably compact. Consider the Gaussian experiment of Theorem 9.1, with uniformly computable preparations, transition matrices $T_{\theta,t}^a$, means $\mu_{\theta,t,j}^a$, and covariance matrices $\Sigma_{\theta,t,a}$. Suppose the same known bounds U, σ_-, σ_+ hold throughout the family. For parameter distance at most s , assume supplied uniform moduli*

$$\|T_{\theta,t}^a - T_{\theta',t}^a\|_{1 \rightarrow 1} \leq \tau_t(s), \quad \|\mu_{\theta,t,j}^a - \mu_{\theta',t,j}^a\|_2 \leq m_t(s), \quad \|\Sigma_{\theta,t,a} - \Sigma_{\theta',t,a}\|_F \leq v_t(s).$$

For categorical Brier prediction, or a uniformly presented bounded terminal task, Theorem 3.1 applies, with the common report reduction (48) and parameter-instrument modulus

$$(17) \quad \omega_t^I(s) = \tau_t(s) + \frac{m_t(s)}{\sigma_-} + \frac{v_t(s)}{\sqrt{2}\sigma_-^2}.$$

Thus the exact minimax risk over all parameter-independent finite-state observers has computably convergent upper and lower certificates at every fixed feasible (n, M) . Transitions may be reducible, deterministic, or permutations; no normalized-filter contraction, positive transition floor or full-rank predictive geometry is assumed here.

Proof. For a normalized column vector x , the instrument is $\text{diag}(\varphi_{\theta,t,j}^a(y))T_{\theta,t}^a x dy$. Subtract first the transition matrix and then the emission density. Integration bounds the first term by $\tau_t(s)$ and the second by the largest L^1 difference of a pair of Gaussian densities. To bound that difference, interpolate the means and covariances on a line segment. Every intermediate covariance has eigenvalues at least σ_-^2 . For a mean direction b , the L^1 norm of the density derivative is

$$\mathbb{E}|b^T \Sigma^{-1}(Y - \mu)| \leq \|b\|_2 / \sigma_-.$$

For a covariance direction C , set $A = \Sigma^{-1/2} C \Sigma^{-1/2}$ and let Z be standard Gaussian. The derivative norm is bounded by

$$\frac{1}{2} \mathbb{E}|Z^T A Z - \text{tr } A| \leq \frac{1}{2} \sqrt{2 \text{tr}(A^2)} \leq \frac{\|C\|_F}{\sqrt{2}\sigma_-^2}.$$

The middle identity follows by orthogonally diagonalizing A and using independence and $\mathbb{E}(Z_i^2 - 1)^2 = 2$. Integrating the two directional bounds proves (17).

All emission mixtures are everywhere positive for every parameter and normalized incoming state. The same box cells, one genuine tail cell, and state- and parameter-independent reconstruction measures therefore have the same support table. The uniform gradient and tail estimates in Theorem 9.1 give (41) without conditioning away the tail. Uniform computability and the displayed moduli give certified cell integrals at every point of a parameter net. Restriction to the future-test effect space gives the parameterwise predictive quotients, even when some states are predictively identical. Categorical Brier loss itself is independent of the inaccessible model, so $\omega^L = 0$ in that case. Preparation variation is charged by ω_0 . The common-programme theorem now applies directly. All memory and timing constraints refer to the one synthesized tuple, not a different filter chosen at each parameter. $\square$

Theorem 4.2 (Unknown noncommuting channels and preparations). *On $\mathbb{C}^D$, with D fixed, consider a computably compact family of instruments*

$$l_{\theta,t}^a(dy)\rho = E_{\theta,y}^{1/2} \mathcal{C}_{\theta,t}^a(\rho) E_{\theta,y}^{1/2} \zeta(dy), \quad \int E_{\theta,y} \zeta(dy) = I,$$

where $\mathcal{C}_{\theta,t}^a$ is completely positive and trace preserving, $\kappa I \preceq E_{\theta,y} \preceq KI$, and ζ is a fixed compact-report reference probability. Assume uniform effective report partitions and instrument oscillation bounds, as in Theorem 9.2 or Proposition 9.3. For nearby parameters suppose

$$\sup_{\rho} \|\mathcal{C}_{\theta,t}^a(\rho) - \mathcal{C}_{\theta',t}^a(\rho)\|_1 \leq c_t(s), \quad \sup_y \|E_{\theta,y} - E_{\theta',y}\|_{\text{op}} \leq e_t(s),$$

with supplied computable moduli. Then Theorem 3.1 applies to every uniformly presented bounded terminal task, in particular Brier prediction of a fixed finite POVM, with

$$(18) \quad \omega_t^I(s) = c_t(s) + \sqrt{K/\kappa} e_t(s).$$

Initial preparations may be pure and channels may be unitary. The effects need not commute. The persistent resource is the classical observer alphabet; the experimental quantum state is not free quantum memory of that observer.

Proof. Replace the channel output first. For a Hermitian difference v , positivity of the instrument and normalization of its integrated trace imply

$$\int \|E_{\theta,y}^{1/2} v E_{\theta,y}^{1/2}\|_1 \zeta(dy) \leq \|v\|_1.$$

Apply the instrument separately to the positive and negative parts of v to obtain this inequality, so the channel contribution is at most $c_t(s)$, not a division by a posterior eigenvalue. For the second replacement, the Sylvester-equation estimate proved in Theorem 9.2 gives $\|E_{\theta,y}^{1/2} - E_{\theta',y}^{1/2}\|_{\text{op}} \leq e_t(s)/(2\sqrt{\kappa})$. Expand the two-factor congruence difference at a density matrix. Its trace norm is at most $\sqrt{K/\kappa} e_t(s)$; integrating proves (18).

Every report density is between κ and K on the same reference support, irrespective of the parameter or purity of the state. Common reconstruction therefore preserves exact supports. Computable positive square roots, channel entries, cell integrals and decision nets give the effective data. Finite-dimensional closure of future-test effects supplies each predictive quotient; indistinguishable decompositions of a density matrix are not retained as different predictive points. Apply Theorem 3.1. For a unitary family, for example, $c_t(s) \leq 2 \sup \|U_{\theta,t}^a - U_{\theta',t}^a\|_{\text{op}}$, by expanding the conjugation difference. No reset or mixing is needed. $\square$

Corollary 4.3 (Singular acquisition with parameter uncertainty). *Suppose a parameter family has a common Cantor reference, uniformly positive instrument densities with a supplied uniform Hölder modulus, and parameter variation bounded as in (13). Ternary cylinders give the same $d_{t,k} \leq L_t 3^{-\alpha k}$ for all parameters. Likewise a common countable support with a computable uniform actual tail bound $T_J \downarrow 0$ gives $d_{t,J} \leq 2T_J$. Separately flagged finite instrument branches may have different output ranks, provided their active probabilities and report supports satisfy the common support condition. In each case (10) holds at unchanged M .*

Proof. Use the same conditional reference on each cylinder or the same tail reconstruction at every parameter. The proof of Proposition 9.3 bounds the positive vector-measure error by the cylinder oscillation or twice the actual tail mass, uniformly. The new parameter modulus is an additional bound on the unnormalized instruments, not a claim about Hausdorff dimension. Flagged branches retained exactly add no binning error. Apply the common-programme reduction and Theorem 3.1.

For an explicit rank-changing family, take the Cantor qubit instrument in Proposition 9.3, allow its coefficient to range over a known compact subinterval of $(0, 1)$, and mix it with separately reported pure- and full-rank preparation branches whose weights range over a compact positive simplex. The branch maps are positive, their weights are bounded away from zero, and their coefficient/weight dependence supplies (13) directly. At the first call, the acquired law is a genuine mixture of an injective affine image of the Cantor law and atoms of different ranks. The original branch masses enter every law and every bound. Subsequent masses are the actual history-dependent ones; no singular stratum is normalized to probability one. This is a raw family with both inaccessible coefficients and changing acquired rank, not merely a declaration of a singular interface. $\square$

4.1. An exact calibration–memory threshold. The following small experiment makes the common-parameter quantifier and the price of retaining acquired calibration information explicit. It is a sharp consequence, not a claim to resolve a general nonlinear filtering problem.

Choose an unknown fixed bit $\vartheta \in \{0, 1\}$ and an unknown fixed noise level $p \in [p_-, p_+]$, where $0 < p_- \leq p_+ < 1/2$. The first paid call reports $C = \vartheta \oplus E$, with $E \sim \text{Bernoulli}(p)$. The second paid call reports an independent fair bit Y . At halt the observer forecasts $X = \vartheta \oplus Y$ under scalar squared loss. A post-decision audit scores that forecast and reveals no information before it is made. Thus there are two observer acquisitions and, if scoring is charged as an acquisition, one additional audit call. There is one action at both phases and the same binary report alphabet without a phase flag. One programme must serve the whole family. This is a finite positive raw kernel with nonconstant acquired calibration information, and all report supports coincide.

Proposition 4.4 (A shared five-state implementation and its sharp converse). *For the preceding task, external widths (m_1, m_2) satisfy*

$$(19) \quad R_{\text{ext}}^{\Theta}(m_1, m_2) = \begin{cases} p_+(1 - p_+), & m_1, m_2 \geq 2, \\ 1/4, & \min(m_1, m_2) = 1. \end{cases}$$

The internally exact two-call class is empty for $M < 3$ and has

$$(20) \quad R_{\text{aut}}^{\Theta}(2, M) = \begin{cases} 1/4, & 3 \leq M \leq 4, \\ p_+(1 - p_+), & M \geq 5. \end{cases}$$

These assertions allow all randomized acquisition and transition rows of the respective classes. If an additional unobserved fixed offset $b \in [-\delta, \delta]$ is added to the audited target, both nonempty risk formulas increase by exactly δ^2 . With the true bit furnished to the programme in advance, the unshifted risk is zero at $M = 4$; furnishing it is a strictly different information pattern.

Proof. For a lower bound even on full-history observers, fix $p = p_+$ and put a uniform prior on ϑ . Given (C, Y) the target equals $C \oplus Y$ with conditional probability $1 - p_+$ and its complement with probability p_+ . Conditional squared-loss projection gives minimum Bayes risk $p_+(1 - p_+)$. A worst-case risk is at least this average. Conversely, retain C and then retain $W = C \oplus Y$. Output p_+ at $W = 0$ and $1 - p_+$ at $W = 1$. For every ϑ and p its risk is

$$(1 - p)p_+^2 + p(1 - p_+)^2 = p_+^2 + p(1 - 2p_+) \leq p_+(1 - p_+).$$

This is one common programme; it proves the upper with two labels at both cuts.

If $m_1 = 1$, the second-call forecast is downstream of Y and fresh randomness, independent of the erased C . Under the uniform prior on ϑ , the target is then a fair bit conditional on the available information, giving risk at least $1/4$. If $m_2 = 1$, the fixed terminal forecast sees neither bit, and X is fair even for each fixed ϑ . The constant $1/2$ attains the bound. Terminal randomization cannot improve it, by convexity of squared loss. These arguments prove (19).

At either phase both reports have positive probability under each model. The common-null-set clock argument of Proposition 7.4 forces the initial, intermediate and stopping label supports to be disjoint. They are also common across parameters, by positivity and the common programme. Their cardinalities therefore obey $1 + m_1 + m_2 \leq M$, giving emptiness below three and a one-label cut when $M \leq 4$. Five states suffice: an initial label s , intermediate labels c_0, c_1 , and stopping labels d_0, d_1 . For the single legal action, the common update and readout are

$$k(c_i | s, y) = \mathbf{1}_{\{i=y\}}, \quad k(d_j | c_i, y) = \mathbf{1}_{\{j=i \oplus y\}}, \quad u(d_0) = p_+, \quad u(d_1) = 1 - p_+.$$

All other occupied-row entries vanish. The labels themselves implement the deadline. The constant-risk three-state programme is a chain of one initial, one intermediate and one stopping label with readout $1/2$. These exact readout values belong to the declared mathematical programme; finite-bit evaluation of a nonrational p_+ is charged by its output precision, not by an assumption that the true p is known. This proves (20), including the converse for all randomized rows.

For the offset extension, place the independent uniform prior on $b \in \{-\delta, \delta\}$ in the lower argument. It adds conditional variance δ^2 , while the one-label lower arguments remain valid. In the upper constructions, the mean forecast error is zero for each ϑ, p , since Y is fair. Thus the cross term with every fixed b vanishes and the risk increases by b^2 , at most δ^2 . For an oracle knowing ϑ , ignore C at a single intermediate state and encode $\vartheta \oplus Y$ in two stopping states, using four states in total. Its zero risk does not give a feasible robust programme. This last comparison explains why the parameter supremum must follow the common-programme choice in (4). $\square$

Corollary 4.5 (Attained calibration and simulation coordinates). *In Proposition 4.4, erase the first report independently with a fixed known probability $a \in (0, 1)$, without making an extra observation. The second report remains the independent fair bit. Give the simulator one physical first-report call and require it to complete the virtual first report before the second report or the postdecision audit. The exact parameter-uniform causal deficiency to the unerased experiment is*

$$(21) \quad \varepsilon = a(1/2 - p_-).$$

For every $M \geq 7$, the worst-case squared prediction risk in this erased experiment, with the same unobserved offset $b \in [-\delta, \delta]$, is exactly

$$(22) \quad R_{\text{aut}}^\Theta(2, M, \delta) = p_+(1 - p_+) + \frac{(1/2 - p_+)^2}{1/2 - p_-} \varepsilon + \delta^2.$$

The optimum uses one parameter-independent seven-state programme. There are two observer acquisitions; an optional scored physical audit is a third call after the decision and does not enlarge the observer deadline. No assertion about the optimal number of labels below seven is made for this enlarged report alphabet.

Proof. On a nonerased first report the simulator copies the bit. On an erasure it supplies a fresh fair bit. It copies the second bit. This is one report-to-report kernel, with no retained information, for all parameters and phases. At a fixed (ϑ, p) , the virtual error probability is $(1 - a)p + a/2$, so its total variation distance from the target first-report law is $a(1/2 - p)$. The second report is independent, and the target scored variable is determined by that report and the fixed parameter. Thus the same bound holds for the complete stopped and scored law. Taking the supremum gives the upper bound in (21).

For the converse fix $p = p_-$ and compare $\vartheta = 0$ and $\vartheta = 1$. The target first-report laws have total variation distance $1 - 2p_-$; the source laws have distance $(1 - a)(1 - 2p_-)$. Every common first-report simulator contracts total variation. If its error in each model is at most e , the triangle inequality therefore gives

$$1 - 2p_- \leq (1 - a)(1 - 2p_-) + 2e.$$

Consequently $e \geq a(1/2 - p_-)$. No second report or audit is available when this virtual report is due. This proves the claimed exact deficiency and explains the importance of the interface.

For a lower risk bound put the uniform prior on ϑ , fix $p = p_+$, and put an independent uniform prior on $b \in \{-\delta, \delta\}$. Even with the complete two-report history, the conditional variance of X is $p_+(1 - p_+)$ when the first bit survives and $1/4$ when it is erased. The offset contributes δ^2 . Hence every programme has worst-case risk at least

$$(1 - a)p_+(1 - p_+) + a/4 + \delta^2 = p_+(1 - p_+) + a(1/2 - p_+)^2 + \delta^2.$$

For attainment, use one initial state, three intermediate states recording 0, 1, *, and three terminal states with readouts $p_+, 1 - p_+, 1/2$. On a surviving bit, update by the same XOR rule as in Proposition 4.4; on an erasure, send either second report to the last terminal state. This is a fixed two-step programme, with all seven states charged. At each fixed (ϑ, p) its risk without the offset is

$$(1 - a)\{p_+^2 + p(1 - 2p_+)\} + a/4.$$

It increases with p and has zero mean forecast error, since the second bit is fair. Every fixed offset therefore adds exactly b^2 . The worst-case risk equals the lower bound. Substitute (21) to obtain (22).

The erasure flag has phase-dependent support, although its support is common across inaccessible parameters at each phase. Thus this construction meets the support-resolved uniform theorem, but the common-null-set clock converse for the binary-report experiment cannot simply be reused to claim a new sharp threshold below seven. The matching risk in (22) comes from a full-history lower bound and an explicit feasible programme, not such a converse. $\square$

5. A CAUSAL ALLOCATION THEOREM

For probabilities ν, λ on a compact convex carrier, write $\nu \preceq_{\text{cx}} \lambda$ when $\int f d\nu \leq \int f d\lambda$ for every continuous convex f . Let $\mathcal{P}_m(K)$ be the probabilities with at most m atoms, not the measures on a chosen grid.

An externally clocked *allocation flow* consists of $\nu_0 = \delta_{q_0}$, $\nu_t \in \mathcal{P}_{m_t}(K_t)$, and a probability $\alpha_t(\cdot | q)$ supported on $\mathcal{A}_t$ at each atom of ν_t , such that

$$(23) \quad \lambda_{t+1} = \int \sum_a \alpha_t(a | q) \Lambda_t(q, a) \nu_t(dq), \quad \nu_{t+1} \preceq_{\text{cx}} \lambda_{t+1} \quad (0 \leq t < n).$$

The measure λ_{t+1} is the actual acquisition mixture proposed by the retained states and actions. It is not the law of a separately run exact filter. Define

$$(24) \quad \Phi_{\text{ext}}(\mathbf{m}) = \inf_{\text{flows (23)}} \left\{ \int g d\nu_n - B_n^* \right\}.$$

All infima below allow arbitrarily close competitors; no unproved attainment assertion is implicit.

For a proposed autonomous label set Z , initial label z_0 , and stopping subset D , put $p_{t,z} \geq 0$, $\sum_z p_{t,z} = 1$, $q_{t,z} \in K_t$, and fix one action probability $\alpha_z(a)$ for every $z \notin D$. Require

$$(25) \quad p_{0,z} = \mathbf{1}_{z=z_0}, \quad q_{0,z_0} = q_0, \quad p_{t,z} = 0 \quad (t < n, z \in D), \quad p_{n,z} = 0 \quad (z \notin D).$$

In addition require $p_{t,z}\alpha_z(a) = 0$ whenever $a \notin \mathcal{A}_t$. All references to feasible occupancies below include this legality constraint. The same label may have different *conditional predictive states* $q_{t,z}$ at different cuts; these are proposed law parameters, not time-varying registers readable by the machine. For each $z \notin D, a$, set

$$(26) \quad \eta_{z,a} = \sum_{t=0}^{n-1} p_{t,z}\alpha_z(a)J_t(q_{t,z}, a), \quad h_{t,z,a} = \frac{d[p_{t,z}\alpha_z(a)J_t(q_{t,z}, a)]}{d\eta_{z,a}}.$$

If $\eta_{z,a} = 0$, its contribution is zero. Otherwise $\sum_t h_{t,z,a} = 1$ almost everywhere. For arbitrary $\phi_{t,j} \in \text{Aff}(K_{t+1})$, define

$$(27) \quad \sum_{t=0}^{n-1} \sum_{j \in Z} p_{t+1,j} \phi_{t,j}(q_{t+1,j}) \leq \sum_{z \notin D} \sum_a \int \max_{j \in Z} \left\{ \sum_{t=0}^{n-1} h_{t,z,a}(y) \phi_{t,j}(F_t(q_{t,z}, a, y)) \right\} \eta_{z,a}(dy).$$

Call the proposed occupancies *simultaneously allocatable* if (27) holds for every such family. Constants among the $\phi_{t,j}$ enforce the masses as well as the barycenters. Finite linear combinations of the determining test coordinates suffice. Define

$$(28) \quad \Phi_{\text{aut}}(n, M) = \inf_{\substack{|Z| \leq M, (25) \\ \text{all inequalities (27)}}} \left\{ \sum_z p_{n,z} g(q_{n,z}) - B_n^* \right\}.$$

No observer transition kernel occurs among these variables.

Theorem 5.1 (Causal acquired geometry and exact/effective resource transfer). *For the prepared instruments and terminal task of Section 2, the following assertions hold.*

(a) *A finite-support externally clocked flow is realizable by a causal observer with the indicated phase alphabets if and only if it satisfies (23). Equivalently, for every step and every collection $\phi_1, \dots, \phi_m \in \text{Aff}(K_{t+1})$, its proposed atoms (p_j, q_j) satisfy*

$$(29) \quad \sum_{j=1}^m p_j \phi_j(q_j) \leq \int \max_{1 \leq j \leq m} \phi_j(w) \lambda_{t+1}(dw).$$

Every failed contraction has a strict witness of this finite polyhedral form.

(b) *Proposed labelled occupancies and action probabilities are realizable by one autonomous tuple of Definition 2.1, stopping exactly at n , if and only if (25) and all simultaneous inequalities (27) hold. Whenever they fail, one finite family of affine tests strictly separates the proposed moments from all phase-independent report allocations.*

(c) *With the same preparation, task and legal actions on both sides,*

$$(30) \quad R_{\text{ext}}(\mathbf{m}) = B_n^* + \Phi_{\text{ext}}(\mathbf{m}), \quad R_{\text{aut}}(n, M) = B_n^* + \Phi_{\text{aut}}(n, M).$$

Every feasible flow is synthesized with exactly its declared labels. The auxiliary predictive states used to specify the flow are programme data, not additional persistent state. The synthesized transition functions need only be measurable; finite-bit or efficient synthesis is not asserted.

(d) *Let ν_t be the retained-state laws of any calibrated observer, and λ_{t+1} its actual pre-pooling law. If the terminal decision is optimal conditional on the retained label, then*

$$(31) \quad R - B_n^* = \sum_{t=0}^{n-1} (A_t + \mathcal{J}_t),$$

where

$$A_t = \mathbb{E} \left[\sum_a \alpha_t(a \mid Z_t) \int V_{t+1}(w) \Lambda_t(Q_t, a)(dw) - V_t(Q_t) \right],$$

$$\mathcal{J}_t = \int V_{t+1} d\nu_{t+1} - \int V_{t+1} d\lambda_{t+1}.$$

Both terms are nonnegative. For an autonomous machine $\alpha_t(\cdot \mid z) = \alpha_z$ is the same kernel at every cut. A suboptimal terminal decision adds its nonnegative conditional decision slack.

(e) *Effective completion. For the additional support-resolved effective presentation of Definition 8.1, the exact optimal value in (c) admits the certified finite intervals (44), with the candidate bound (43). They are uniform over every legal Borel programme of the fixed cardinality. The minimizing listed programme is executable at that same cardinality, with separately charged cell access, readout,*

probability precision, scratch and programme costs. This effective assertion does not apply to arbitrary compact Borel presentations merely by compactness. Its proof is Theorem 8.2.

Parts (a)–(d) do not assume a density, a stable update, finite-dimensional closure, or full task curvature. Part (e) has the additional effective positive-instrument presentation stated there. It is an exact geometric feasibility and optimization theorem on the declared compact predictive class, not a universal numerical formula for arbitrary Borel experiments. Its two dual tests differ: in (27) the maximum is taken *after summing over phases*. Moving that maximum inside the phase sum would allow a different programme at every unobserved phase and would be a strictly weaker test.

Corollary 5.2 (Exact and approximate completion). *A zero-excess observer exists with a specified interface and budget if and only if its corresponding feasible allocation flow has $A_t = \mathcal{J}_t = 0$ at every cut and zero terminal decision slack. For every $\epsilon > 0$, an observer with excess at most ϵ exists if and only if a feasible flow has $\sum_t (A_t + \mathcal{J}_t) \leq \epsilon$. The infimal excess is zero if and only if such flows exist for every positive ϵ .*

Proof. Apply the realization parts of Theorem 5.1 and the sum of nonnegative terms in (31). The last statement separates an infimum of zero from attained exact sufficiency. $\square$

For example, a piecewise-affine Bayes risk can have zero Jensen loss on a nontrivial cell. Its acquired dimension alone cannot force a positive lower. At the other extreme, Section 10 estimates the same Jensen defects by actual cell mass and distributional curvature.

Proposition 5.3 (Checkpoint identity). *For a full-history policy π , let ν_n^π be its actual terminal predictive law and $B_n^\pi = \int g d\nu_n^\pi$. Then*

$$(32) \quad R_{\text{cp}}(n, M) = \inf_{\pi} \{B_n^\pi + D_M^g(\nu_n^\pi)\}, \quad D_M^g(\lambda) = \inf_{\substack{\nu \in \mathcal{P}_M(K_n) \\ \nu \preceq_{\text{cx}} \lambda}} \left\{ \int g d\nu - \int g d\lambda \right\}.$$

In particular, the two policy-dependent terms are not minimized separately.

Proof. Conditional on a final label, affinity gives the conditional predictive barycenter and the best risk g at that barycenter. Necessity and sufficiency of the terminal contraction follow from the one-step allocation lemma below. Full-history independent randomization may be conditioned out for this checkpoint relaxation; alternatively include its kernels in π and the displayed infimum is unchanged. For fixed finitely many readouts, assigning each predictive state to a least conditional loss readout is Borel and no worse than a randomized assignment. Optimizing the readouts therefore also gives the equivalent partition definition of D_M^g . This purification concerns a terminal encoder, not a stationary finite controller. $\square$

The affine components in the displayed duals are continuous functions on the predictive carrier. Lemma 2.2 identifies their precise relation to executable tests and uniform test-span limits. The finite family of maxima in a global obstruction is not generally one scored continuation. The dominating measure $\eta_{z,a}$ is a finite measure of aggregate occupancy-action mass $\sum_t p_{t,z} \alpha_z(a)$; it need not be a probability or even have mass at most one. The same action-indexed report space makes its shared row well-typed across phases.

6. BARYCENTRIC ALLOCATION AND THE DEPLOYED LAW

We give the finite-target form of the classical convex-order argument explicitly, since it is also the building block for phase-compatible implementation.

Lemma 6.1 (Finite barycentric allocation). *Let K be compact metrizable and convex as above, $\lambda \in \mathcal{P}(K)$, and $\nu = \sum_{j=1}^m p_j \delta_{q_j}$. The following are equivalent:*

- (1) *there are Borel functions $r_j : K \rightarrow [0, 1]$, $\sum_j r_j = 1$ λ -almost everywhere, such that*
- $$(33) \quad \int r_j d\lambda = p_j, \quad \int r_j(w) f(w) \lambda(dw) = p_j f(q_j) \quad (f \in \text{Aff}(K));$$
- (2) *all inequalities (29) hold;*
- (3) $\nu \preceq_{\text{cx}} \lambda$.

When the inequalities fail, finitely many determining affine coordinates and m affine expressions provide a strict witness. Atomic acquired laws are allowed, and randomization is not silently replaced by deterministic cells.

Proof. Consider the simplex of allocations $r = (r_1, \dots, r_m)$ in $(L^\infty(\lambda))^m$. It is weak-star compact: it is contained in the unit ball, and positivity and the identity $\sum_j r_j = 1$ are weak-star closed. Map it to all its masses and affine-coordinate moments. Each coordinate is an integral against an $L^1(\lambda)$ function, so the image is compact and convex in a product of real lines. If the target moments $(p_j, p_j q_j)$ are outside that image, separation in this locally convex product gives a continuous linear functional involving only finitely many coordinates. Combining the coordinates for each j yields an affine ϕ_j . The supremum of the separated functional over allocations is exactly

$$\int \max_j \phi_j(w) \lambda(dw) :$$

a measurable least maximizing index attains it. This proves equivalence of the first two assertions, including strict separation. A weak-star allocation has Borel versions on the standard Borel space; modify on one null set to keep it in the simplex everywhere. Matching the separating coordinates matches the barycenter, hence every continuous affine function.

Given (33), generate $W \sim \lambda$ and choose label j with probability $r_j(W)$. Then $\mathbb{E}[W \mid j] = q_j$, where conditional expectation is understood through affine coordinates. Jensen's inequality gives $\nu \preceq_{\text{cx}} \lambda$. Conversely, for any affine family, convex order gives

$$\sum_j p_j \phi_j(q_j) \leq \int \max_i \phi_i d\nu \leq \int \max_i \phi_i d\lambda,$$

which is assertion two. This also proves sufficiency without invoking a separate infinite-dimensional transport theorem. The argument is the finite-target martingale/convex-order mechanism of Strassen[17], not a new static comparison theorem. $\square$

Lemma 6.2 (Calibration under all incoming reports). *Suppose label i has probability p_i and conditional predictive state q_i . Draw its action from α_i , execute the raw report kernel, and let $W = F_t(q_i, a, Y)$. If a next-label rule satisfies $\mathbb{E}[W \mid Z' = j] = q'_j$, then q'_j is the actual predictive state conditional on $Z' = j$. In particular this applies to the kernel $r_j(W)$ of Lemma 6.1 under the full acquired mixture $\lambda = \sum_{i,a} p_i \alpha_i(a) \Lambda_t(q_i, a)$.*

Proof. The incoming label represents a conditional mixture of physical histories, not a physical preparation performed anew. Equation (2) and predictive sufficiency imply that the conditional expectation of each determining future test, given (i, a, Y) , is its affine evaluation at W . Integrating this identity over *all* incoming labels and reports sent to j gives its evaluation at q'_j . The tests determine the predictive quotient. This is precisely the conditional state under the deployed observer. There is no comparison with the full-history posterior on a different controlled path. $\square$

Proof of Theorem 5.1(a), (c) for the external interface, and (d). For necessity, at each persistent cut of any observer take its actual conditional predictive states. Their weighted law is ν_t , with support size at most $|Z_t|$. If several labels have the same state, average their action distributions using their conditional label probabilities. This gives the same acquired mixture λ_{t+1} . The next retained state is a conditional barycenter of the pre-pooling state, so Jensen's inequality gives

(23). This step records only marginals needed for subsequent predictive evolution; it does not assert equality of the original labelled trajectories after duplicate states are merged.

For sufficiency, assume a flow. Use its atoms as phase labels and its action probabilities as the programmed action kernel. Lemma 6.1 provides $r_{t+1,j}$ for $\lambda_{t+1} \rightarrow \nu_{t+1}$. Programme

$$k_t(j \mid i, a, y) = r_{t+1,j}(F_t(q_{t,i}, a, y)).$$

The conditional-state parameters are constants in this phase's programme, not a persistently updated real vector. Lemma 6.2 proves the prescribed occupancies and states by induction from q_0 . Thus each feasible flow yields a legal observer of risk $\int g d\nu_n$. Conversely, replacing any observer's final readout by the best conditional readout gives this risk. Infimizing proves the first equality in (30).

For the telescope, let Q_t denote the actual retained conditional state. Before pooling at the next cut,

$$\mathbb{E}V_{t+1}(W_{t+1}) - \mathbb{E}V_t(Q_t) = A_t \geq 0$$

by the Bellman minimum. Calibration and concavity give

$$\mathbb{E}V_{t+1}(Q_{t+1}) - \mathbb{E}V_{t+1}(W_{t+1}) = \mathcal{J}_t \geq 0.$$

Adding and telescoping yields (31). The argument uses the machine's actual action distribution and remains valid at nonsmooth action switches. It neither compares independently optimized shadow controllers nor assumes Bellman-optimal actions for a stationary machine that reuses a label. $\square$

Remark 6.3 (Why the dual is operational). An arbitrary terminal splitting of a preparation is not automatically obtainable from a fixed sensor. The right side of (29) is evaluated under that sensor's actual λ , not under an unconstrained information-design experiment. Likewise, local contractions at different cuts do not authorize phase-dependent code. The simultaneous test below imposes precisely the shared-kernel restriction that ordinary one-step convex order omits.

6.1. Finite obstructions to a global risk target. For the externally clocked problem, compactness upgrades the witness for one proposed flow to a finite obstruction valid for every flow at an unattainable risk level. This is not an interchange of an infimum with a pointwise dual supremum.

Theorem 6.4 (Compact completion and finite obstructions). *Under the compactness and weak-continuity hypotheses of Theorem 5.1, every finite external width profile has an optimal allocation flow and an optimal measurable observer. For each phase fix a countable family $h_{t,1}, h_{t,2}, \dots$ dense in the continuous convex functions on K_t , each a finite maximum of finite affine combinations of the determining coordinates. Let c_k be the minimum terminal risk subject to the atomic-width, action-marginal and acquisition constraints of (23), but retaining only the first k convex-order inequalities at each cut. Then*

$$(34) \quad c_k \uparrow R_{\text{ext}}(\mathbf{m}).$$

Consequently, if a number $r < R_{\text{ext}}(\mathbf{m})$ is proposed as a feasible risk, finitely many of these acquired-law inequalities, together with the resource and risk constraints, already exclude every proposed flow with risk at most r . This is a finite family of tests, not necessarily a single test that defeats every candidate. No effective bound on its size or computability is asserted for general compact Borel data.

Proof. Represent the action assignments by probabilities γ_t on $K_t \times \mathcal{A}_t$ whose first marginal is ν_t . The sets of probabilities with at most m_t atoms are compact: a sequence of such probabilities has convergent subsequences of weights and atom locations after padding by zero weights. Marginal constraints are closed. The map $\gamma_t \mapsto \lambda_{t+1} = \int \Lambda_t(q, a) \gamma_t(dq, da)$ is weakly continuous by weak

continuity of each Λ_t and finiteness of the action set. Thus the candidate space before imposing convex order is compact. Every convex-order inequality is closed, and terminal risk is continuous. Feasibility is nonempty: choose one legal action at every cut and replace each acquired law by its single barycenter. Hence a risk minimizer exists and is synthesized by Theorem 5.1.

For completeness, the required countable polyhedral family exists. The closed convex epigraph of a continuous convex function on compact K_t can be separated from each point lying strictly below it by a continuous affine functional of the ambient countable product and the height coordinate. The height coefficient cannot vanish, because the epigraph contains a point with the same base coordinate. The resulting affine minorant uses only finitely many coordinates. A finite subcover of K_t gives a finite maximum approximating the function uniformly from below. Approximating its finitely many coefficients by rationals gives a countable uniformly dense collection, with constants included. Inequalities for this collection imply all continuous convex inequalities by uniform approximation.

The finite-constraint feasible sets are nonempty, compact and nested, so their minima c_k are nondecreasing and at most the full optimum. Choose minimizers and a convergent subsequence in the compact candidate space. For every fixed test index, all sufficiently late members obey its inequality, and the limit does too. Thus the limit satisfies all convex-order constraints, while its terminal risk is $\lim_k c_k$. This proves (34). If r is strictly below that limit, some finite k has $c_k > r$, yielding the asserted obstruction. Its proof does not bound k and does not establish a finite computational representation of the carrier or kernels. $\square$

Remark 6.5 (Why compactness supplies no enumeration rate). Consider a fair hidden bit, a constant report, one acquisition and two allowed terminal labels, under binary error loss. The acquired predictive law is $\delta_{1/2}$ and the true risk is $1/2$. The proposed retained law $(\delta_0 + \delta_1)/2$ has the same barycenter and risk zero. It passes every affine inequality but fails the convex-order condition. A countable dense convex-test enumeration may start with any prescribed finite number of affine functions. The corresponding finite relaxation therefore remains at zero for an arbitrarily long initial segment. Density and compactness alone give no bound on that index. This example concerns the supplied enumeration, not a lower bound for every algorithm. The effective theorem obtains its quantitative certificate from additional instrument and computation data, not by reinterpreting topological compactness as an effective modulus.

For clarity, the candidate space in the compactness argument may be written using $\gamma_t \in \mathcal{P}(K_t \times \mathcal{A}_t)$ with first marginal ν_t supported on at most m_t points, initial $\nu_0 = \delta_{q_0}$, and $\lambda_{t+1} = \int \Lambda_t(q, a) \gamma_t(dq, da)$. If h_i enumerates the convex test family, its k th closed candidate subset imposes

$$\int h_i d\nu_{t+1} \leq \int h_i d\lambda_{t+1} \quad (t < n, i \leq k), \quad c_k = \min \left\{ \int g d\nu_n : \gamma \text{ in that subset} \right\}.$$

All finite atoms and action weights are chosen jointly for this fixed experiment. No measurable dependence of an optimizer on an external model parameter is asserted. The moment-image topology is the product of the real mass and affine-coordinate lines, so a continuous linear separator depends on finitely many coordinates. Completed-probability measurable allocation functions have Borel versions on the standard-Borel carrier, with values on null sets completed arbitrarily.

7. ONE PROGRAMME ACROSS ALL PHASES

Lemma 7.1 (Simultaneous report allocation). *For fixed $(p_{t,z}, q_{t,z}, \alpha_z)$ satisfying (25), the inequalities (27) are equivalent to the existence of Borel probabilities $k(j \mid z, a, y)$, independent of t ,*

such that, for every $t < n$, label j and $f \in \text{Aff}(K_{t+1})$,

$$(35) \quad p_{t+1,j} f(q_{t+1,j}) = \sum_{z \notin D, a} p_{t,z} \alpha_z(a) \int k(j \mid z, a, y) f(F_t(q_{t,z}, a, y)) J_t(q_{t,z}, a)(dy).$$

The case $f = 1$ is included.

Proof. For each (z, a) use the finite dominating measure $\eta_{z,a}$ in (26). In the finite product of spaces $(L^\infty(\eta_{z,a}))^{|Z|}$, take the weak-star compact convex set of simplex-valued functions. The map to the right sides of (35) is continuous into the product of affine-coordinate moments: its test functions are $h_{t,z,a}(y) f(F_t(q_{t,z}, a, y))$, which are in $L^1(\eta_{z,a})$. If the target moments are not in this compact convex image, finite-coordinate separation gives a family $\phi_{t,j}$. For each incoming pair (z, a) the support function of its allocation simplex is

$$\int \max_j \sum_t h_{t,z,a}(y) \phi_{t,j}(F_t(q_{t,z}, a, y)) \eta_{z,a}(dy).$$

Summing over (z, a) is exactly the right side of (27). A Borel least maximizing index realizes this support function. Separation proves both directions and gives a strict finite witness of infeasibility. Borel versions and zero-measure pairs are treated as in Lemma 6.1. $\square$

Proof of Theorem 5.1(b) and the autonomous part of (c). An actual autonomous tuple supplies its cut occupancies and conditional states. Its single transition kernel satisfies (35) by conditioning on the incoming label, action and report. The stop rule gives (25). Thus the simultaneous inequalities are necessary.

Conversely, Lemma 7.1 constructs one kernel $k(j \mid z, a, y)$ for all phases. Use it with the proposed, phase-independent α_z . Starting from z_0 , induction in (35) gives exactly $p_{t,z}$ and $q_{t,z}$, by the all-incoming calibration argument. The induction does not require the machine to know $q_{t,z}$ or t : those objects appear only in its offline feasibility certificate, while the executed kernel is the single k . The zero occupancies in (25) ensure that it never reaches a stopping label before n and reaches one at n almost surely. Choose a best readout at each positive-mass terminal label. This gives the stated risk and uses only Z . Taking infima proves the second equality in (30). $\square$

Remark 7.2 (A strict distinction between the tests). Suppose incoming label 1 and a fixed action see the same deterministic report in two phases, and both next predictive carriers consist of a singleton. A proposal sends that report to label 1 with probability one in the first phase and to label 2 with probability one in the second. Each separate barycentric allocation is feasible. Set $\phi_{0,1} = \phi_{1,2} = 1$ and the other two constants to zero, with equal incoming phase masses. The left side of the simultaneous inequality is 2, and its right side is 1. Thus no single kernel implements the proposal. The obstruction is not a failed geometric covering; it is incompatible reuse of one programme location.

7.1. Finite matrices and a sharp programme-reuse threshold. For finite reports, write $\mathbf{f}_t(q)$ for the constant and a finite affine coordinate basis, when the carrier is finite-dimensional. Given the proposed occupancies, predictive states and action rows, put

$$A_{t,z,a,y} = p_{t,z} \alpha_z(a) J_t(q_{t,z}, a)(y) \mathbf{f}_{t+1}(F_t(q_{t,z}, a, y)), \quad b_{t,j} = p_{t+1,j} \mathbf{f}_{t+1}(q_{t+1,j}).$$

Then the simultaneous allocation is precisely the finite feasibility system

$$(36) \quad \sum_{z,a,y} A_{t,z,a,y} k_{z,a,y,j} = b_{t,j}, \quad k_{z,a,y,j} \geq 0, \quad \sum_j k_{z,a,y,j} = 1.$$

It is a linear system for a *fixed proposal*; optimizing the occupancies and action rows is not generally one linear programme. Finite-dimensional linear-programming duality gives the finite version of (27). Each report row occurs once, even if used at several phases. With finite raw states,

retaining the unnormalized physical occupancy vectors instead yields polynomial recursions in the controller rows, directly connecting this formulation with finite-state-controller parameter synthesis[23].

A small complete experiment shows both nontrivial feasible reuse and a sharp risk threshold. There are four paid calls and action types *sense* and *respond*. At calls one and three only sense is legal; its binary reports have respective laws $(3/4, 1/4)$ and $(1/4, 3/4)$. At calls two and four only responses 0, 1 are legal. Call two returns a deterministic *return* marker and call four a deterministic *finish* marker. A hidden scoring register records whether the first response was 0 and the second response was 1. After the observer has halted, a fixed audit scores the average of the two errors; the scoring register and audit outcome are never readable before the decision. The audit adds one physical scoring call if included in acquisition totals, not one more controller transition. This is an ordinary positive finite kernel after adjoining the phase and score to the hidden state.

Theorem 7.3 (A finite sharp reuse frontier). *For this prepared four-call experiment, with one autonomous programme and the entire label set counted,*

$$(37) \quad R_{\text{aut}}(4, M) = \begin{cases} +\infty, & M < 3, \\ 1/2, & M = 3, \\ 1/4, & M = 4, \\ 0, & M \geq 5. \end{cases}$$

An external observer with one label at every phase has risk zero. The internally timed thresholds do not follow from a clock-neutral-report theorem: the return and finish markers have different phase supports.

Proof. First suppose one sense location is reused. If a, b are the probabilities that its report rule ultimately selects response zero on reports zero and one, the two phase probabilities are

$$u = (3a + b)/4, \quad v = (a + 3b)/4.$$

Every $(u, v) \in [0, 1]^2$ is separately phasewise feasible. One common row exists exactly on the nontrivial parallelogram

$$(38) \quad 0 \leq 3u - v \leq 2, \quad 0 \leq 3v - u \leq 2, \quad a = (3u - v)/2, \quad b = (3v - u)/2.$$

These are the finite simultaneous constraints. In particular $u - v \leq 1/2$, so the average error $(1 - u + v)/2$ is at least $1/4$. The common deterministic row $a = 1, b = 0$ attains it. For an interior feasible example, $a = 3/4, b = 1/4$ gives $(u, v) = (5/8, 3/8)$; the displayed inverse recovers the same randomized row at both phases.

Labels reached at sense and response cuts cannot coincide: their stationary action distributions would have to be supported in two disjoint legal action sets. A stopping label is different from both. Thus at least three labels are necessary. With exactly three, there is one response label, whose fixed response distribution is the same at both scored calls; its average error is $1/2$. A chain using the markers to return to the sense label and then halt attains it.

With four labels, if there is only one response label its average error is still $1/2$. Otherwise there is exactly one sense label and at most two response labels. Any randomization at the response labels can be composed with the sense row to give the common binary probabilities a, b above, so the same $1/4$ lower applies to *all* randomized programmes. Attain it with labels s, r_0, r_1, d : sense at s , send report y to r_y , respond j at r_j , send return to s and finish to the stopping label d . These are phase-independent rules.

With five labels, use two sense locations s_0, s_1 , two response locations and one halt location. Initially sense at s_0 and send either report to r_0 ; return goes to s_1 , whose reports both go to r_1 ;

finish goes to halt. The risk is zero. The external observer knows the stage and can choose the correct response without retaining the reports, also at zero risk. No unscored internal memory has been used. $\square$

Proposition 7.4 (Exact conversion of timing conventions). *For every experiment in the main theorem,*

$$(39) \quad R_{\text{ext}}(n, M) \leq R_{\text{aut}}(n, M), \quad R_{\text{aut}}(n, 1 + \sum_{t=1}^n m_t) \leq R_{\text{ext}}(\mathbf{m}).$$

For the second inequality the source observer's phase label is included in the autonomous label. Suppose additionally that, for each action a , every $J_t(q, a)$ is mutually absolutely continuous with one common probability ζ_a on reports, uniformly in the sense of null sets over t, q . Then every internally exact- n observer has disjoint positive-probability label supports at all $n + 1$ cuts, and

$$(40) \quad R_{\text{aut}}(n, M) = \inf_{\substack{m_t \geq 1 \\ 1 + \sum_{t=1}^n m_t \leq M}} R_{\text{ext}}(\mathbf{m}).$$

Its class is nonempty exactly when $M \geq n + 1$. No uniform likelihood-ratio lower is required.

Proof. An external observer may ignore its clock and run an autonomous one, proving the first inequality. Conversely, attach t to each phase label of an external observer; its initial label is separate and its final labels stop. This gives a stationary tuple on the disjoint union and the second inequality.

For disjointness, suppose a positive-mass label z occurs at cuts $s < t$. Starting from that label, run the same stationary action and transition kernels against reference reports ζ_a and the same fresh randomization. For every finite continuation length, successive conditioning shows equivalence between actual and reference trace laws: the next-report density is positive and finite almost everywhere, and randomization is unchanged. Integrating over any incoming conditional physical state or history preserves the same null sets. From cut t , stopping after $n - t$ further calls has probability one, hence reference probability one. Starting from the reused label at cut s , the reference machine is identical, so the same event has actual probability one. It stops before n , a contradiction. The case $t = n$ follows directly from the stopping set. Thus the cut supports are disjoint. Their sizes obey $1 + \sum_t m_t \leq M$, giving the reverse inequality in (40). A chain with n legal calls and one terminal label proves nonemptiness at $n + 1$, irrespective of its quality of prediction. $\square$

Report time stamps or a freely supplied stage index invalidate the disjointness argument, not the simultaneous allocation theorem. An autonomous machine with such reports is still governed by (27); it may be able to reuse labels. The conversion result therefore specifies which timing cost is a resource convention and which compatibility constraint survives that convention.

In the reference-trace argument, every action with positive probability at a reused label is legal at both occurrences by the occupancy legality constraint. Equivalence of trace null sets is not a uniform likelihood-ratio estimate; no such stronger bound is used.

8. AN EFFECTIVE FRONTIER AT A FIXED PERSISTENT BUDGET

The exact allocation invariant retains an entire law-valued trajectory. We now prove that, for an effectively presented class of positive instruments, its value admits certified finite approximations without increasing the observer's persistent alphabet. No stability of the normalized filter is assumed. The assertion is stronger than pointwise approximation of a fixed controller: its lower bound is uniform over all admissible Borel controllers of the declared size.

8.1. The experimental approximation certificate. Let V be a finite-dimensional real vector space with a closed generating pointed cone V_+ and a strictly positive linear mass functional e . Assume the base $\mathcal{B} = \{x \in V_+ : e(x) = 1\}$ is compact. Use the base norm

$$\|v\|_{\mathcal{B}} = \inf_{v=v_+-v_-, v_{\pm} \in V_+} \{e(v_+) + e(v_-)\}.$$

A raw instrument $\mathfrak{l}_t^a$ is a countably additive measure of positive linear maps on V , with $e\mathfrak{l}_t^a(\mathcal{Y}_a) = e$. Preparation is $x_0 \in \mathcal{B}$. The terminal loss is $L_u(x)$, where $0 \leq L_u \leq He$ in the dual order. The compact decision set is U . Classical substochastic matrices and finite-dimensional completely positive quantum instruments are two instances, with ℓ^1 and trace norm respectively. The physical state is not a register supplied to the observer.

Finite words of unnormalized instruments and terminal effects generate a finite-dimensional space of executable-test expectations. Restricting physical preparations to that space gives the compact affine predictive quotient. Weak continuity of the post-report law also follows in this finite-dimensional setting. Choose an interior normalized positive vector x_* ; compactness of the base gives $x \preceq Cx_*$ for every normalized x . The report measure at x_* dominates all vector instrument entries. For a continuous bounded function of the outgoing predictive state, its integrand is the report density times that function. It is continuous in x wherever the density is positive, extends continuously by zero when the density vanishes, and is dominated by C times the reference density. Dominated convergence proves continuity on the physical base. Representative independence and the compact quotient map descend it to the predictive carrier. Thus the exact theorem applies; the effective argument below uses the unnormalized instrument identities directly. A redundant physical representation may be used for offline calculation. Its dimension is not substituted for the dimension of an acquired law.

Definition 8.1 (Support-resolved effective presentation). For each refinement index h , each action has a finite Borel report partition $\mathcal{C}_a^h = (C_{a,b})_{b=1}^{B_a}$ and probability measures $R_{a,b}$ supported on their cells. The partition and reconstruction measures are independent of phase and physical state. They come with a finite support table $S_{t,a} \subseteq \{1, \dots, B_a\}$ such that, for every $x \in \mathcal{B}$,

$$e\mathfrak{l}_t^a(\cdot)x|_{C_{a,b}} \text{ is equivalent to } R_{a,b} \quad (b \in S_{t,a}), \quad \mathfrak{l}_t^a(C_{a,b}) = 0 \quad (b \notin S_{t,a}).$$

Equivalence concerns null sets, not a uniform likelihood-ratio bound. Cells inactive at every legal phase may be removed. Define the regenerated instrument

$$\widehat{\mathfrak{l}}_t^a(dy) = \sum_b R_{a,b}(dy)\mathfrak{l}_t^a(C_{a,b}).$$

There are supplied computable bounds $d_{t,h} \geq 0$ such that

$$(41) \quad \sup_{a \in \mathcal{A}_t, x \in \mathcal{B}} \int \|(\mathfrak{l}_t^a - \widehat{\mathfrak{l}}_t^a)(dy)x\|_{\mathcal{B}} \leq d_{t,h}, \quad \sum_{t < n} d_{t,h} \longrightarrow 0.$$

The integral is the total variation norm of a finite-dimensional vector measure. Matrix entries of each cell instrument, preparation, and terminal loss at a given decision are computable, with certified absolute-error enclosures. For every $\rho > 0$ a finite, explicitly encoded set U_ρ is supplied such that every $u \in U$ has $v \in U_\rho$ with

$$\sup_{x \in \mathcal{B}} |L_u(x) - L_v(x)| \leq \rho.$$

Cell membership is an executable operation of declared description, scratch and time cost; alternatively an implementation provides a separate probability-of-misbinning certificate and preserves the report support signature. Nothing in computability of integrals alone supplies a bounded-cost analogue comparator.

The support table permits phase flags and other clock-revealing reports. It need not be constant in t . Its role is to preserve *exact* legality and stopping while eliminating the continuum report variable from a stationary programme.

Put $A = |\mathcal{A}|$, $B_\Sigma = \sum_a B_a$, and, for a dyadic denominator $Q = 2^b$, set

$$(42) \quad P = MA + M^2 B_\Sigma, \quad E_h = H \min\{1, \sum_{t < n} d_{t,h}\}, \quad \beta_Q = H \min\{1, n(A + M)/Q\} + \rho.$$

Let $\mathcal{G}_{h,Q,\rho}$ contain all M -label autonomous programmes whose action and cell-to-label rows have denominator Q and whose readouts lie in U_ρ , retaining only the programmes accepted by the finite support-graph check below. Write $K_\rho = |U_\rho|$. Unused labels and arbitrary rows at unreachable labels are allowed.

Theorem 8.2 (Uniform finite certificates for the exact frontier). *Fix $n \geq 1$ and $M \geq 1$ in a support-resolved effective experiment. Admissibility of each member of $\mathcal{G}_{h,Q,\rho}$ is decidable by finite graph reachability. The Borel autonomous class is empty if and only if this finite set is empty, for any supplied partition, any dyadic $Q \geq 1$, and nonempty U_ρ . Moreover*

$$(43) \quad |\mathcal{G}_{h,Q,\rho}| \leq M 2^M K_\rho^M (Q + 1)^P.$$

For each retained programme g , compute an interval $[l_g, u_g]$ containing its actual risk, of length at most $\eta > 0$. If the class is nonempty, put $l = \min_g l_g$ and $u = \min_g u_g$. Then

$$(44) \quad \boxed{l - E_h - \beta_Q \leq R_{\text{aut}}(n, M) \leq u, \quad u - l \leq \eta.}$$

A programme attaining $\min_g u_g$ is implementable on the original experiment at the same M , with risk at most u and therefore excess over the optimal M -label Borel risk at most $E_h + \beta_Q + \eta$. The bounds may be clipped to $[0, H]$. They give an effective convergent two-sided approximation of $B_n^* + \Phi_{\text{aut}}(n, M)$.

For an external width profile the same assertion holds, with phase-indexed rows and

$$(45) \quad P_{\text{ext}} = \sum_{t < n} \left(m_t |\mathcal{A}_t| + m_t m_{t+1} \sum_{a \in \mathcal{A}_t} B_a \right), \quad m_0 = 1,$$

and count bound $K_\rho^{m_n} (Q + 1)^{P_{\text{ext}}}$. In β_Q replace $n(A + M)$ by $\sum_{t < n} (|\mathcal{A}_t| + m_{t+1})$. An externally supplied phase remains free only in this second model.

8.2. Proof: common reconstruction before discretizing programmes.

Lemma 8.3 (Report reduction preserves one programme). *Let R_h be the optimum when the reports are replaced by their cell indices, keeping the physical cell instruments $\mathbf{l}_t^a(C_{a,b})$ and the same observer interface. Under Definition 8.1,*

$$(46) \quad R \leq R_h \leq R + E_h.$$

Both transformations preserve the entire persistent alphabet, all positive-probability action legality constraints, and the exact deadline.

Proof. Any cell programme runs on the original reports by computing their cells; its joint physical-state/label law is exactly that given by the cell instruments. This proves $R \leq R_h$.

Conversely, let $k(j \mid z, a, y)$ be any admissible stationary Borel programme. Define the single cell row

$$\bar{k}(j \mid z, a, b) = \int k(j \mid z, a, y) R_{a,b}(dy).$$

This is independent of phase. Running k against the regenerated instrument has exactly the same state/label law as running $\bar{k}$ against the cell instrument. The reconstruction is an offline averaging identity, not a persistent report simulator or an exact posterior computation.

For the error, at any cut the joint state/label law is a family (v_z) of positive vectors with $\sum_z e(v_z) = 1$. A positive instrument followed by an action row and a report-to-label row is a contraction on the direct-sum base norm. Indeed decompose each signed input into positive parts; positivity and preservation of total mass bound the sum of output norms by the sum of input norms. On positive normalized inputs, replacing the instrument changes that sum by at most $d_{t,h}$, by (41) and the stochastic row sums. Successive replacements therefore change the final joint state/label vector by at most $\sum_t d_{t,h}$. A terminal effect in $[0, He]$ changes its expected loss by at most H times this norm, also at most H . This proves the risk error E_h . The argument is valid for quantum instruments without postulating a hidden classical trajectory for noncommuting states.

It remains to check the exact interface. For an active cell, an edge $z \rightarrow j$ after action a is possible under the fine programme precisely when $\int k(j \mid z, a, y) dR_{a,b} > 0$. This equivalence holds at every active phase and every incoming positive conditional physical state. Finite paths have positive probability exactly when their action edges and active-cell edges have positive probability, by induction. Thus the original programme and its cell average have the same possible label/action paths at each phase. Inactive cells cannot occur in either. Premature halt, non-halt at n , and illegal actions are consequently absent from one programme if and only if they are absent from the other. Taking infima proves (46), also for classes whose optimum is not attained. The proof for external rows is identical with t retained as a permitted argument. $\square$

Lemma 8.4 (Finite support graphs and dyadic approximation). *For the cell experiment, a stationary programme is legal and stops exactly at n if and only if the following finite recursion accepts it. Start with the initial label. At every $t < n$ reject a reachable stopping label or a positive action outside $\mathcal{A}_t$; otherwise follow all positive row entries for all $b \in S_{t,a}$. At n require every reachable label to stop. For every admissible real-valued cell programme there is one in $\mathcal{G}_{h,Q,\rho}$ with risk increase at most β_Q .*

Proof. The support equivalence and positivity induction in Lemma 8.3 prove the graph criterion. No infinite-horizon recurrence is being tested. For a stochastic row of length r , take the floors of Q times its entries and distribute the remaining units only to entries in its original support. The resulting row has the same or smaller support, total mass one, and total variation at most r/Q . This construction works even at $Q = 1$; it does not require a lower bound on nonzero row probabilities. Apply it once to each shared stationary row, not separately at each phase. Removing edges cannot introduce a previously forbidden finite path, so legality and exact halting are preserved. There remain outgoing edges at every nonterminal row because its mass is one.

Couple the action and next-label choices up to their first discrepancy. At one call their total mismatch probability is at most $(A + M)/Q$. Conditional on equality so far, the physical experiment and label histories are the same. A union bound over n calls bounds the loss increase by $H \min(1, n(A + M)/Q)$. Selecting a loss- ρ -net readout at each stopping label adds at most ρ . The same proof gives the displayed external sum. This comparison is uniform over the original row probabilities, including arbitrarily small positive entries. $\square$

Proof of Theorem 8.2. Enumerate initial labels, stopping subsets, readouts, and numerator rows. There are at most M , 2^M , and K_ρ^M choices for the first three. A length- r denominator- Q row has at most $(Q + 1)^r$ possibilities; summing the row lengths gives P . Apply the support check to each. If a Borel observer exists, Lemma 8.3 and then Lemma 8.4 give a member of this finite list. The converse follows by execution, proving the empty-class assertion.

Let R_G be the least actual risk in the retained finite list. The preceding lemmas give

$$R \leq R_G \leq R + E_h + \beta_Q.$$

Each listed programme's risk is a finite sum of compositions of computable cell instruments, row probabilities and terminal effects. Certified interval arithmetic computes it to any prescribed

absolute accuracy; computability here includes the oracle moduli in Definition 8.1, not just existence of the integrals. Hence $l \leq R_G \leq u$. If i minimizes l_i , then $u \leq u_i \leq l_i + \eta = l + \eta$, proving the interval width. A programme minimizing u_g has risk at most u . The error in (44) follows. All steps for the external interface use the stated phase widths. Finally choose the computable refinement, Q , loss net and evaluation tolerance to make E_h , β_Q and η tend to zero. This is an effective value approximation, not an assertion that a general optimal Borel programme is attained or has a finite exact description. $\square$

Corollary 8.5 (Finite global obstruction and computation of excess). *For the effective class and any strict target $r < R_{\text{aut}}(n, M)$, refinement eventually produces $l - E_h - \beta_Q > r$. The finite list of admissible programmes, its certified risk intervals, the support graph and the displayed error bounds form a global obstruction with candidate count bounded by (43). This is not a bound on the number of affine tests in an arbitrary enumeration in Theorem 6.4.*

The same construction on finite cell-history trees computes B_n^ : after report reduction there are finitely many action/report histories of depth at most n , and deterministic contingent actions and loss-net readouts suffice. Thus Φ_{aut} and Φ_{ext} have certified intervals obtained by subtracting a baseline interval for the same experiment. The policy-dependent acquisition and compression terms in (32) are not minimized separately.*

Proof. Choose total interval error smaller than $R - r$. For the baseline, report reduction applies to full-history controllers with the external history retained; the coarsened history alphabet is finite. At a finite decision tree each action randomization can be purified backwards because the conditional objective is linear in that row. Replace terminal decisions by U_ρ . Finite enumeration and the same error calculation yield a baseline interval. Subtraction gives $[R_L - B_U, R_U - B_L]$, with the lower clipped at zero for the excess. This last computation may use vastly larger offline history trees; it does not grant extra state to the deployed M -label observer. $\square$

8.3. Programme, numerical and physical resources. A listed programme has a direct read-only encoding of length

$$(47) \quad O(P \log(Q + 1) + M \log(K_\rho + 1) + M + \log(M + 1)) + C_{\text{cell}} + C_{\text{out}}.$$

The supplied physical model, its integration algorithms, and a fixed interpreter are not included in this programme count. C_{cell} is the cell-classifier description and C_{out} encodes the finite decision net. They are not zero by convention. A dyadic row uses exactly b fresh bits and a cumulative row scan; no rejection loop or retained random seed is required. In addition to the $\lceil \log_2 M \rceil$ persistent bits, transition scratch is

$$O(b + \log(M + 1) + \log(A + 1) + \log(1 + \max_a B_a)) + W_{\text{cell}} + W_{\text{out}}.$$

Apart from physical acquisitions, the two scans use $O(n(A + M)(b + 1))$ bit operations, plus cell and readout costs. Offline search inspects at most (43) candidates, each with a finite support check and a certified risk evaluation. No polynomial-time or optimal programme-length claim follows.

If numerical cell evaluation changes a report cell with total probability at most χ along the ideal deployed law, and preserves its support signature, first-discrepancy comparison adds at most $H \min(1, \chi)$ to the attained upper risk. Signature preservation retains the exact deadline. A readout error contributes its own loss modulus. A common-task simulator of defect ε adds $H\varepsilon$ and multiplies persistent state by its certified simulator alphabet, as in Proposition 13.1. These are separate costs; they do not alter the ideal lower bound by assuming that every allowed numerical tolerance is attained.

9. EFFECTIVE RAW REALIZATIONS WITHOUT FILTER CONTRACTION

The next results verify the same effective theorem from two different raw instruments. They do not assume an already available filter register, a Fisher matrix, or a smooth acquired density. The stronger geometric rate hypotheses used later are not required here.

Theorem 9.1 (Untruncated Gaussian experiments). *Let the hidden state be finite, the action set finite, and let a call apply a known stochastic matrix T_t^a before reporting $Y \mid X^t = j \sim N(\mu_{t,j}^a, \Sigma_{t,a})$ in $\mathbb{R}^r$. Suppose all entries are computable with certified enclosures, $\|\mu_{t,j}^a\|_\infty \leq U$, and covariance eigenvalues lie in $[\sigma_-^2, \sigma_+^2]$, $\sigma_- > 0$. The matrices T_t^a may have zeros, be reducible, or be permutations; no mixing, rank or normalized-update contraction condition is imposed. Use categorical Brier prediction, or any effectively presented bounded terminal task.*

For $R > U$, partition $[-R, R]^r$ into cubes of side at most h , and use one additional cell for its entire complement. This is a support-resolved effective presentation with

$$(48) \quad \begin{aligned} B_a &\leq \lceil 2R/h \rceil^r + 1, & G &= (2\pi)^{-r/2} \sigma_-^{-(r+1)} e^{-1/2}, \\ d_{t,h} &\leq G\sqrt{r}(2R)^r h + 4r \exp\{-(R-U)^2/(2\sigma_+^2)\}. \end{aligned}$$

Consequently Theorem 8.2 computes the optimal fixed-label external or internally timed risk with certified finite upper and lower bounds, for every fixed horizon and feasible budget.

Proof. On positive column vectors, write $\mathbf{l}_t^a(dy)x = \text{diag}(\varphi_{t,j}^a(y))T_t^a x dy$. This is positive and mass preserving after integration. Every mixture report density is positive everywhere, even when a transition is deterministic. All cells are therefore active at every legal phase. Use normalized Lebesgue measure on each bounded cell, and one fixed nonsingular Gaussian conditioned on the complement as the reconstruction there. Neither measure depends on t or on the hidden state.

For a Gaussian density, writing $z = \Sigma^{-1/2}(y - \mu)$ gives

$$\|\nabla \varphi(y)\|_2 \leq (2\pi)^{-r/2} \sigma_-^{-(r+1)} |z| e^{-|z|^2/2} \leq G.$$

The difference between a density and its cell average is at most $G\sqrt{r}h$ on a bounded cell. Integrating over the box gives the first term of (48). A positive normalized input gives a convex mixture of these scalar bounds, so no hidden-state cardinality factor is needed. On the complement, the total variation norm between the original vector measure and its reconstructed version is at most twice its mass. A coordinate union bound and the Gaussian Chernoff bound give complement mass at most $2r \exp(-(R-U)^2/(2\sigma_+^2))$. This proves the second term. The tail is retained as a paid report cell; it is not discarded by conditioning.

For example, to make $H \sum_t d_{t,h} \leq \xi$, take $R > U$ sufficiently large that the tail term is at most $\xi/(2nH)$, and then $h \leq \xi/(2nHG\sqrt{r}(2R)^r)$. Gaussian box integrals have effective certified approximations by bounded-domain quadrature with derivative bounds and tail control. Composition with the computable stochastic matrices preserves computability. The Brier decision simplex has explicit rational nets: a denominator- D simplex rounding changes the expected Brier loss by at most $8s/D$ for s categories, since its ℓ^1 change is at most $2s/D$ and the loss modulus is at most four times that change. Thus all computational clauses of Definition 8.1 hold for ideal cell access.

There is no need to identify the full hidden simplex with the predictive quotient. Restriction to the invariant span of future-test effects gives the exact compact quotient, including experiments with indistinguishable hidden states. Its actually acquired laws determine the invariant in Theorem 5.1. The matrices are an offline representation, not a free state available to the deployed observer. $\square$

For numerical reports $\tilde{Y}$ with $\|Y - \tilde{Y}\|_\infty \leq b$ outside a separately charged exceptional event, any cell change lies within b of a grid boundary. If there are at most $q + 1$ boundaries per

coordinate, the Gaussian marginal density bound gives per-call probability at most

$$\min\{1, 2r(q+1)b/(\sqrt{2\pi}\sigma_-)\}, \quad q = \lceil 2R/h \rceil.$$

This includes the box boundary and applies to all conditional mixtures under every deployed policy. Every cell has the same support signature, so such errors cannot create an illegal stopping path. The reporter's precision and physical comparison costs remain separate from the controller table.

Theorem 9.2 (Noncommuting instruments with no reset requirement). *Let $\mathcal{H} = \mathbb{C}^D$, with fixed finite D , and let $\mathcal{C}_t^a$ be any computably given completely positive trace-preserving maps. On the report cube $[-1, 1]^r$ with uniform probability measure, take*

$$E_y = I + s \sum_{i=1}^r y_i H_i, \quad \mathfrak{l}_t^a(dy)\rho = E_y^{1/2} \mathcal{C}_t^a(\rho) E_y^{1/2} 2^{-r} dy,$$

where the Hermitian matrices are computable, $s \geq 0$ is computable, and $\kappa I \preceq E_y \preceq KI$ with a known $\kappa > 0$. The effects need not commute and the channels may be unitary. For a fixed finite POVM and its Brier task, Theorem 8.2 applies with

$$(49) \quad B_a \leq \lceil 2/h \rceil^r, \quad d_{t,h} \leq \sqrt{K/\kappa} s \left(\sum_i \|H_i\|_{\text{op}}^2 \right)^{1/2} \sqrt{r} h.$$

No positive lower bound on the evolving state's eigenvalues is needed for this conclusion. Constants depend on the fixed dimension and displayed instrument bounds.

Proof. Because the cube is centered, $\int E_y 2^{-r} dy = I$, proving normalization. The report density is at least κ on every normalized positive input, so all cells are active; uniform conditional cube measure is a common reconstruction. For positive definite $A, B \succeq \kappa I$, the equation

$$\sqrt{A}(\sqrt{A} - \sqrt{B}) + (\sqrt{A} - \sqrt{B})\sqrt{B} = A - B$$

has the integral solution obtained by integrating $e^{-t\sqrt{A}}(A - B)e^{-t\sqrt{B}}$ over $t \geq 0$. It yields $\|\sqrt{A} - \sqrt{B}\|_{\text{op}} \leq \|A - B\|_{\text{op}}/(2\sqrt{\kappa})$. For a density matrix σ , the two-factor difference estimate therefore gives

$$\|E_y^{1/2} \sigma E_y^{1/2} - E_v^{1/2} \sigma E_v^{1/2}\|_1 \leq \sqrt{K/\kappa} \|E_y - E_v\|_{\text{op}}.$$

Cauchy-Schwarz bounds the last norm by $s(\sum_i \|H_i\|_{\text{op}}^2)^{1/2} |y - v|$. Average over a cell and then over the probability measure on the cube to obtain (49). The channel output is always a density matrix, so its purity causes no denominator problem in this unnormalized estimate.

The square root is uniformly computable on the positive spectral interval with a gap κ ; integration of the continuous matrix entries over rational boxes is consequently effective with supplied moduli. The fixed-POVM probabilities are affine in ρ , and the Brier loss has computable finite decision nets as above. Closure of finite-dimensional future-test effects produces the predictive quotient. No infinite-dimensional operator statement and no optimal quantum-memory claim are implicit: the persistent resource here is the classical controller alphabet, while the physical quantum system belongs to the experiment. $\square$

Proposition 9.3 (Singular reports, atomic tails and changing rank). *The effective theorem also holds in the following two classes, under its support and computability clauses.*

(i) *Reports have a prescribed Cantor probability reference and positive instrument density $A_t^a(y)$. If, on normalized positive inputs, $\|(A_t^a(y) - A_t^a(v))x\|_{\mathcal{B}} \leq L_t |y - v|^\alpha$, then the 2^k ternary-cylinder partition gives $d_{t,k} \leq L_t 3^{-\alpha k}$. Any fixed finite family of separately flagged positive instruments can be retained exactly; their outputs may have different ranks.*

(ii) *Reports are countable, with the same positive support at every phase and input. If the actual probability of reports outside the first J is uniformly at most a supplied computable $T_J \downarrow 0$, preserve the first J reports and pool the tail. Then $B = J + 1$ and $d_{t,J} \leq 2T_J$, with no regularity of the tail updates.*

Proof. In (i), reconstruct from the reference conditioned on each cylinder. Its diameter is 3^{-k} , so the density oscillation, averaged over the reference probability, is at most $L_t 3^{-\alpha k}$. A flagged discrete outcome is a singleton cell and incurs no error. In (ii), the vector measures agree on the retained atoms. On the tail, positivity bounds the norm of each of the original and reconstructed measures by its actual mass. The triangle inequality gives $2T_J$. Choose a computable full-support conditional reference on that tail. Common support ensures that pooling does not invent a phase-specific possible path.

For a concrete singular acquired law, take a qubit initially in $I/2$, the identity channel, the Cantor reference on $[0, 1]$, and $E_y = I + s(y - 1/2) \text{diag}(1, -1)$ with $0 < s < 1$. Its mean effect is I , report density is one at that preparation, and the acquired posterior is exactly $E_y/2$. This injective affine image of the Cantor law is singular in the trace-one affine space. The instrument is not a re-preparation of its input. Separately flag a fixed positive fraction of calls which prepare a pure state, and another fixed fraction which prepare $I/2$; the remaining fraction uses this continuous instrument. The flags have input-independent positive mass. At the first call the actual acquired law has a singular continuous component with its original mixture weight and atoms of different rank. Later laws use their actual history-dependent mixture weights, and repeated flags allow rank changes in both directions. Refinement never normalizes away the continuous or atomic masses. The preceding estimates prove an effective frontier for this experiment; no dimension-only power law or normalized-filter contraction is asserted. $\square$

10. QUANTITATIVE ACQUIRED GEOMETRY

The exact criterion separates implementability from estimates on the objective. We now re-prove a regular quantitative consequence, retaining the all-incoming and clock hypotheses explicitly. This section's curvature and integer-allocation mechanism is inherited from the preceding article[14]; the exact criterion above is not inferred from it.

All carriers in this section are one compact convex $K \subset \mathbb{R}^d$ with nonempty affine interior. Fix constants comparing the chosen norm $\|\cdot\|$ to the Euclidean norm $|\cdot|$. Suppose every $\Lambda_t(q, a)$ has density at most ρ relative to the displayed d -dimensional affine volume, for *all* q, a, t . Suppose g is Lipschitz and κ -strongly concave in Euclidean coordinates, $\kappa > 0$. Assume J_t is uniformly TV-continuous and

$$(50) \quad W_1(\Lambda_t(q, a), \Lambda_t(q', a)) \leq \beta_t \|q - q'\|, \quad 0 < \beta_t \leq B, \quad \prod_{j=t}^{t+b-1} \beta_j \leq \theta < 1$$

for every complete block of a fixed length b . The same experimental constants are used for every horizon. Zero coefficients can be enlarged to positive sufficient coefficients; no claim of sharpness for β_t is made.

Theorem 10.1 (Regular acquired-geometry transfer). *Under these hypotheses there are constants $0 < c \leq C < \infty$, depending only on K, d , the norm comparison, $\rho, \kappa, \text{Lip}(g), B, b, \theta$, such that, for every $n \geq 1$ and $M \geq 1$,*

$$(51) \quad cM^{-2/d} \leq R_{\text{cp}}(n, M) - B_n^* \leq R_{\text{ext}}(n, M) - B_n^* \leq CM^{-2/d}.$$

If report laws also have the common null sets of Proposition 7.4, then the autonomous class is nonempty exactly when $M \geq n + 1$ and

$$(52) \quad c(M - n)^{-2/d} \leq R_{\text{aut}}(n, M) - B_n^* \leq C(M - n)^{-2/d}.$$

All legal adaptive acquisition policies are included. Without the block-product assumption the same construction gives, with $M \geq n + 1$ in the autonomous expression,

$$(53) \quad \begin{aligned} R_{\text{ext}}(n, M) - B_n^* &\leq C_0 M^{-p} \sum_{t=1}^n a_t, \\ R_{\text{aut}}(n, M) - B_n^* &\leq C_0 (M - n)^{-p} \left(\sum_{t=1}^n a_t^{1/(1+p)} \right)^{1+p}, \\ p &= 2/d, \quad a_t = \text{Lip}(g) \prod_{j=t}^{n-1} \beta_j. \end{aligned}$$

The autonomous upper does not need clock-neutral reports; that hypothesis is used only for its matching lower.

For a finite measure λ and a partition $\mathcal{P}$, set

$$\mathcal{J}_v(\lambda, \mathcal{P}) = \sum_{C \in \mathcal{P}} \lambda(C) v(\bar{\lambda}_C) - \int v d\lambda.$$

Zero-mass cells are omitted. This is the Jensen objective of the corresponding barycentric contraction. The following estimate includes singular acquired measures, although a clean power law needs an additional mass estimate.

Lemma 10.2 (Boundary-valid acquired-mass curvature estimate). *Let $K \subset \mathbb{R}^d$ be compact, convex, and have nonempty interior. If v is concave and Euclidean L -Lipschitz on K , then $-v$ has a globally convex L -Lipschitz extension f . For every finite measure λ on K and grid side $h > 0$,*

$$(54) \quad \mathcal{J}_v(\lambda, \mathcal{P}_h) \leq C_d h^{2-d} \sum_C \lambda(C) \mu_f(\bar{B}_{3\sqrt{d}h}(z_C)).$$

If λ is a probability law with density at most ρ , then for every integer $L_0 \geq 1$ there is a Borel partition with at most L_0 cells such that

$$(55) \quad \mathcal{J}_v(\lambda, \mathcal{P}) \leq C_{K,d} \rho L L_0^{-2/d}.$$

Neither an interior support margin nor differentiability of v is required.

Proof. Put $f_0 = -v$. At each $x \in \text{int } K$ select all supporting slopes of f_0 . Their Euclidean norms are at most L : directional difference quotients can be taken in every direction at an interior point. Take the supremum, on all $\mathbb{R}^d$, of the resulting supporting affine functions. This supremum is finite, convex, L -Lipschitz and agrees with f_0 on $\text{int } K$; continuity gives equality on K . In particular, no assertion about arbitrary Lipschitz extensions preserving convexity is being used.

First suppose the extension is smooth. For a cell center z , subtract its tangent plane and call the resulting convex function w . Thus $w \geq 0$ and $w(z) = 0$. Put $R = 2\sqrt{d}h$. For any x in the cell, $B_{R/2}(x) \subset B_R(z)$. Writing av for the normalized spherical average, the submean inequality for w , radial monotonicity, and the divergence theorem give

$$w(x) \leq \frac{C_d}{R^d} \int_{B_R(z)} w \leq C_d \text{av}_{\partial B_R(z)} w \leq C_d R \text{av}_{\partial B_R(z)} \partial_r w = C_d R^{2-d} \int_{B_R(z)} \Delta f.$$

Here radial monotonicity gives $\int_{B_R} w \leq (R/d) \int_{\partial B_R} w$. The radial inequality is $w(z + R\omega) \leq R \partial_r w(z + R\omega)$. This argument also applies in dimension one, with the boundary integral

interpreted as the sum over two endpoints. If b_C is the conditional barycenter, the affine tangent terms cancel and

$$\int_C f d\lambda - \lambda(C)f(b_C) = \int_C w d\lambda - \lambda(C)w(b_C) \leq \lambda(C) \sup_C w.$$

Mollify f on $\mathbb{R}^d$. The convex mollifications converge uniformly on bounded sets, and their positive Laplacian measures converge weakly to μ_f . The limsup of the mass of the radius- R ball is bounded by the mass of the closed ball of radius $3\sqrt{d}h$. Summing the finitely many relevant cells proves (54).

A Lipschitz convex function has uniformly bounded Laplacian mass on any fixed bounded region: choose a smooth nonnegative cutoff χ that equals one there and use

$$\int \chi d\mu_f = - \int \nabla \chi \cdot \nabla f \leq L \int |\nabla \chi|.$$

For h bounded above, the enlarged cell balls have bounded overlap, and $\lambda(C) \leq \rho h^d$. Equation (54) therefore gives $C_{K,d}\rho L h^2$. Enclose K in a fixed cube and subdivide each coordinate into $\lfloor L_0^{1/d} \rfloor$ intervals. There are at most L_0 cells. Since $\lfloor L_0^{1/d} \rfloor \geq L_0^{1/d}/2$, one has $h \leq C_K L_0^{-1/d}$. This proves (55), including $L_0 = 1$, by changing the fixed constant. The extension is used only in this proof, not as an observer state. $\square$

Lemma 10.3 (Integer allocation with a baseline label). *Let $a_t \geq 0$, $p > 0$, and $m \geq 1$. There are integers $L_t \geq 1$ with $\sum_t L_t \leq n + m - 1$ and*

$$(56) \quad \sum_t a_t L_t^{-p} \leq 2^p m^{-p} \left(\sum_t a_t^{1/(1+p)} \right)^{1+p}.$$

When all $a_t = 0$ choose $L_t = 1$. When $m = 1$ the same choice is valid.

Proof. Otherwise put $w_t = a_t^{1/(1+p)}$, $A = \sum_t w_t$ and $L_t = 1 + \lfloor (m-1)w_t/A \rfloor$. The total count is as stated. For $m = 1$, $\sum_t a_t \leq A^{1+p}$. For $m \geq 2$ and $w_t > 0$, $L_t \geq (m-1)w_t/A$ gives $\sum_t a_t L_t^{-p} \leq A^{1+p}(m-1)^{-p} \leq 2^p A^{1+p} m^{-p}$. Zero-weight phases still have one real label. $\square$

Proof of Theorem 10.1. At each phase the action set is common to all incoming states. The Wasserstein bound therefore implies $\text{Lip}(V_t) \leq \beta_t \text{Lip}(V_{t+1})$, by comparing the same action at two states before taking the finite minimum. Thus $\text{Lip}(V_t) \leq a_t$. Concavity was established independently from fixed-programme affine risks.

Starting from the preparation, choose a Bellman-minimizing action at every current label, partition the *full incoming mixture* at the next cut, and use its cell barycenters. Lemma 6.2 makes this a calibrated machine under its own law. Every pre-pooling law obeys the density bound, since the bound holds at all incoming mixture states and actions. Lemma 10.2 with L_t cells and the exact identity (31) give $R - B_n^* \leq C_0 \sum_t a_t L_t^{-p}$. Choosing $L_t = M$ proves the external weighted upper. Choosing the integers of Lemma 10.3 with $m = M - n$ and using disjoint phase labels proves the autonomous weighted upper with total count $1 + \sum_t L_t \leq M$. No exact filter is executed outside these labels.

For $k = n - t$, block decomposition yields the explicit estimate

$$a_t \leq \text{Lip}(g) \max(1, B^{b-1}) \theta^{\lfloor k/b \rfloor}.$$

Hence both sums in (53) are uniformly bounded. This proves the two horizon-uniform uppers, including $m = 1$.

For the lower, condition on the full history immediately before the final action/report. Every conditional final predictive law has density at most ρ , so its mixture under any randomized

adaptive policy does too. If S is this full-history state and a final label Z takes at most L values, its best forecast has conditional state $Q = \mathbb{E}[S \mid Z]$. Strong concavity implies

$$\mathbb{E}g(Q) - \mathbb{E}g(S) \geq \frac{\kappa}{2} \mathbb{E}|S - Q|^2.$$

The L balls of radius $r = (2\rho v_d L)^{-1/d}$ about the possible values of Q cover mass at most $1/2$. Thus $\mathbb{E}|S - Q|^2 \geq r^2/2$. This also covers a randomized label assignment: Q still has at most L values. Since the full-history risk of this policy is at least B_n^* , the same lower holds above B_n^* . Use $L = M$ for checkpoint and external observers. For an autonomous observer, Proposition 7.4 gives $L \leq M - n$. This completes the proof. The lower concerns acquired mass under every admissible policy, not a packing of a reachable set. $\square$

Proposition 10.4 (Finite strata with unnormalized mass). *Replace the density hypothesis by a finite report-readable decomposition of each $\Lambda_t(q, a)$ into submeasures supported on fixed compact convex affine sets $K_i \subset K$, of dimensions $d_i \leq r$, $r \geq 1$. Positive-dimensional submeasures have densities at most ρ_i in their own affine volumes; zero-dimensional types are atoms. Suppose every full-history terminal law has an r -dimensional submeasure of actual mass at least $w > 0$ and density at most ρ_* . Retain the task and block hypotheses on K . Then (51)–(52) hold with exponent $2/r$. The lower constant retains a factor*

$$(57) \quad \kappa w^{1+2/r} \rho_*^{-2/r}.$$

The autonomous lower again requires common report null sets.

Proof. For a phase allocation $L \geq 2s$, where s is the number of types, assign one cell to every atomic type and at least $\lfloor L/s \rfloor$ cells to each positive-dimensional type, reducing counts if necessary so the total is at most L . Restrict the curvature lemma to each K_i and each unnormalized component. Since $d_i \leq r$, its loss is at most a fixed multiple of $\text{Lip}(V_t)L^{-2/r}$. Type is available from the current report, so the pair (type, cell) is an executable label rule. All incoming edges of that pair are pooled together. The resulting barycenter is in K_i . For $L < 2s$, one global cell costs at most $\text{Lip}(V_t) \text{diam}(K)$; enlarge the constant by $(2s)^{2/r}$. Integer allocation and block summation proceed as before.

For the lower choose radius $r_0 = (w/(2\rho_* v_r L))^{1/r}$. The union of L ambient balls intersects the affine component in volume at most $L v_r r_0^r$, and therefore covers at most $w/2$ of that submeasure. The remaining mass contributes at least $(w/2)r_0^2$ to squared distortion. Strong concavity gives (57). No conditional normalization removes w , and no mass from other strata is subtracted. $\square$

The ambient Euclidean chart in the curvature estimate is the finite-dimensional affine span of K ; the chosen predictive norm is compared to its Euclidean norm with fixed constants. The curvature measure μ_f is distinct from the task's strong-concavity constant κ . In Theorem 10.1, B_n^* remains horizon- and task-dependent. Uniformity in n concerns the excess constants, not a common stationary learning curve across different terminal tasks. Optimization over adaptive acquisitions is within the known-kernel, finite-action, phase-legal interface and its uniform hypotheses.

11. TWO RAW-KERNEL REALIZATIONS AND CONVENTIONAL TASKS

These verifications are consequences of the general theorems, never their premises. Their kernel mechanisms were established in the preceding article[14]; we include complete proofs and specify ordinary prediction tasks.

Lemma 11.1 (Positive-instrument coupling). *Let states be probability vectors in ℓ^1 or finite-dimensional density matrices in trace norm. Let C be a positive trace-preserving affine preparation channel with contraction coefficient γ . Let A_y be a positive linear instrument density with*

$\int A_y \zeta(dy)$ trace preserving. For $q = C(s)$ put $u_y = A_y(q)$, $r_y = \text{tr } u_y$, and $F_y = u_y/r_y$ on positive-mass reports. Then

$$(58) \quad W_1(\text{Law } F_Y, \text{Law } F'_{Y'}) \leq 3\gamma \|s - s'\|, \quad \text{TV}(\text{Law } Y, \text{Law } Y') \leq \frac{\gamma}{2} \|s - s'\|.$$

Proof. The positive and negative parts of a signed vector or Hermitian matrix v give

$$\int \|A_y(v)\| \zeta(dy) \leq \|v\|, \quad \int |\text{tr } A_y(v)| \zeta(dy) \leq \|v\|.$$

Couple common report mass $\min(r_y, r'_y)$ at the same y . There

$$\min(r_y, r'_y) \|F_y - F'_y\| \leq \|u_y - u'_y\| + |r_y - r'_y|.$$

The state space has diameter at most two, so the remaining report mass costs at most $\int |r_y - r'_y| d\zeta$. Summing proves the Wasserstein bound; the second integral directly gives the TV bound. These are sufficient coefficients, not claimed optimal ones. $\square$

11.1. Gaussian hidden-state sensing. There are $d + 1$ hidden states, a fixed finite action family, and known stochastic matrices T_t^a with every entry at least $\tau > 0$. A call applies T_t^a and then reports

$$Y \mid X' = j, a \sim N(\mu_j^a, \Sigma_a), \quad Y \in \mathbb{R}^{m_{\text{obs}}}, \quad m_{\text{obs}} \geq d.$$

The parameter set is compact: the means are uniformly bounded, the eigenvalues of Σ_a are uniformly bounded above and below, and the least eigenvalue of $B_a \Sigma_a B_a^T$ is uniformly positive, where B_a has rows $(\mu_j^a - \mu_0^a)^T \Sigma_a^{-1}$. At each phase at least one legal T_t^a is invertible. Put $\gamma_t = \max_a \frac{1}{2} \max_{i,j} \sum_k |T_t^a(i, k) - T_t^a(j, k)|$.

Theorem 11.2 (Categorical filtering). *If $\beta_t = 3\gamma_t$ obeys (50), all conclusions of Theorem 10.1 hold for the conventional loss*

$$L(X_n, u) = \|e_{X_n} - u\|_2^2, \quad u \in \Delta_d,$$

with predictive carrier Δ_d . The full-history baseline is the usual optimal categorical filtering Bayes risk in this controlled experiment. Constants are uniform in n, M on the stated parameter class. Every Gaussian report, including its tails, is a paid acquisition.

Proof. For incoming p set $q = pT_t^a$; then $\tau \leq q_j \leq 1 - d\tau$. With the Gaussian emission densities f_j , the raw joint kernel gives

$$r(q, y) = \sum_j q_j f_j(y), \quad u_j = F_j(p, a, y) = q_j f_j(y) / r(q, y).$$

Unnormalized finite continuation probabilities are linear in p , proving the instrument mixture identities on the whole simplex. All report densities are positive on $\mathbb{R}^{m_{\text{obs}}}$, and therefore equivalent to one fixed nonsingular Gaussian probability reference for each action, at every phase.

The actual posterior law has uniformly bounded density even at the simplex boundary. Indeed its logits satisfy

$$z_j = \log(u_j/u_0) = (B_a y)_j + (c_a)_j + \log(q_j/q_0), \quad (c_a)_j = -\frac{1}{2}(\mu_j^T \Sigma_a^{-1} \mu_j - \mu_0^T \Sigma_a^{-1} \mu_0).$$

Their law is a finite mixture of nonsingular d -dimensional Gaussians, with uniform bounds on means and covariance eigenvalues. Hence its density obeys $h_q(z) \leq C \exp(-c|z|^2 + C|z|)$. The softmax map has Jacobian $\prod_{j=0}^d u_j$, and

$$(\prod_j u_j)^{-1} = (1 + \sum_{j=1}^d e^{z_j})^{d+1} e^{-\sum_j z_j} \leq C_d e^{C_d |z|}.$$

Consequently $h_q(\text{logit } u) / \prod_j u_j$ is bounded uniformly on the entire simplex interior. The boundary has zero acquired mass; no clipping or conditioning was used.

The row coefficient gives $\|pT - p'T\|_1 \leq \gamma_t \|p - p'\|_1$: decompose the difference into equal positive and negative masses and bound the distance of two row mixtures. Lemma 11.1 proves report continuity and the required Wasserstein estimate. This also gives weak continuity of the acquired kernel.

For predictive separation, distinct Gaussian means give linearly independent emission densities: divide a purported relation by the common quadratic Gaussian factor and restrict the resulting exponentials to a line with distinct slopes. Successive limits force all coefficients to vanish. The next-report law thus identifies q , and an invertible legal T_t^a identifies p . Conversely p determines all future tests, so this is the predictive quotient rather than a redundant parameterization.

Finally $\ell(p, u) = 1 - 2p \cdot u + \|u\|_2^2$ is affine in p and has Bayes risk $g(p) = 1 - \|p\|_2^2$. It is Lipschitz and strongly concave on the affine simplex. Apply Theorem 10.1. The proof optimizes over the full declared adaptive policy class; it does not infer a strict feedback advantage merely from the existence of several sensors. $\square$

For instance, two-state invertible transition matrices with row coefficients repeating 0.8, 0.8, 0.02 have individual certified coefficients as large as 2.4, but every three-step product is $0.3456 < 1$. Weak mixing by itself is not being substituted for this explicit block condition.

11.2. Noncommuting finite-dimensional instruments. Let $D \geq 2$, $d = D^2 - 1$, and take an orthonormal real basis $H_1, \dots, H_d$ of traceless Hermitian matrices. Choose $s > 0$ with $s \sum_i \|H_i\|_{\text{op}} < 1$. With uniform probability measure on the report cube $[-1, 1]^d$, put

$$E_y = I + s \sum_i y_i H_i, \quad \mathcal{I}_{t,y}^a(\rho) = E_y^{1/2} \mathcal{C}_t^a(\rho) E_y^{1/2},$$

where

$$\mathcal{C}_t^a(\rho) = (1 - \gamma_t^a) \tau_t^a + \gamma_t^a U_a \rho U_a^*, \quad 0 < \gamma_- \leq \gamma_t^a \leq \gamma_+ < 1, \quad \tau_t^a \succeq \tau I.$$

The affine channel extends linearly by multiplying its first term by $\text{tr } \rho$. It is a paid mixture of preparation and a unitary channel. Since $\int E_y 2^{-d} dy = I$, the instrument is positive and normalized. No common eigenbasis of the effects is assumed.

Theorem 11.3 (Prediction of an informationally complete measurement). *If $\beta_t = 3 \max_a \gamma_t^a$ satisfies the block condition, Theorem 10.1 holds on the density-matrix carrier for Brier prediction of the outcome of a fixed finite informationally complete POVM. Affine volume means Lebesgue volume in the displayed orthonormal traceless chart, and the Wasserstein metric uses trace norm. The constants need not be uniform in D . The terminal measurement is paid when included in the physical acquisition count.*

Proof. Let $q = \mathcal{C}_t^a(\rho)$, $r_y = \text{tr}(E_y q)$ and $\sigma_y = E_y^{1/2} q E_y^{1/2} / r_y$. Uniform positive lower bounds on q, E_y give equivalent report supports on the whole cube. Linearity of the unnormalized quantum instruments proves mixture closure. The coupling lemma supplies the Wasserstein coefficient $3\gamma_t^a$ and TV continuity.

For the density calculation, at positive definite q, σ define

$$A = q^{-1/2} (q^{1/2} \sigma q^{1/2})^{1/2} q^{-1/2}.$$

This is the unique positive definite solution of $AqA = \sigma$. The inverse of the normalized congruence map on the trace- D positive hyperplane is

$$(59) \quad E = \frac{DA^2}{\text{tr } A^2}.$$

Since A is positive definite, $E^{1/2} = \sqrt{D / \text{tr } A^2} A$, proving the substitution in both directions. The map and its inverse are smooth; their differentials on the trace hyperplanes are inverse. The affine map $y \mapsto E_y$ has full rank. The permitted q and closed cube lie in a compact subset of

the positive domains, so the absolute Jacobian of $y \mapsto \sigma_y$ has a positive uniform lower bound. The bounded report density and one-to-one change of variables prove a uniform acquired-density upper. Cube boundaries are null, and no multiplicity factor is hidden.

The report density $1 + s \sum_i y_i \text{tr}(H_i q)$ identifies all traceless coordinates of q ; the affine channel is injective since $\gamma_t^a > 0$. Thus the density matrix is exactly the predictive quotient for the declared tests. It is derived from the raw instruments, not supplied to the finite observer as a register.

For a concrete fixed POVM take $G_{i,\pm} = (I \pm eH_i)/(2d)$ with $0 < e < 1/\max_i \|H_i\|_{\text{op}}$. These effects sum to I . Its outcome probabilities $P_{i,\pm}(\rho) = \text{tr}(G_{i,\pm}\rho)$ form an injective affine map. Under ordinary Brier loss the Bayes risk is $1 - \|P(\rho)\|_2^2$. Its strong-concavity constant is twice the squared least singular value of the linear part of P on the traceless chart. This is a probability-forecast task for an actual fixed measurement, not an assertion that every discrimination loss has full-dimensional curvature. Theorem 10.1 now applies. $\square$

As a finite-stratum extension, replace a fixed fraction $\vartheta < 1$ of calls by a flagged preparation of one of finitely many pure states and use the displayed continuous instrument otherwise. A subsequent positive channel sends pure states back into the interior. The type is visible in the current report; the continuous terminal submeasure has its actual mass $1 - \vartheta$. Provided the corresponding coupling block bound holds, Proposition 10.4 applies with its unnormalized mass factor. This verifies genuine rank changes without asserting that arbitrary or unobserved quantum strata satisfy that proposition.

In the Gaussian realization the operative nondegeneracy is $\text{rank } B_a = d$, including when the observation dimension is larger than d . The invertible legal transition used to separate predictive states may depend on phase; action availability belongs to the phase-specific predictive interface and is not a clock that an autonomous programme may read.

For the quantum inverse, one can see the full differential directly. With $K = (q^{1/2}\sigma q^{1/2})^{1/2}$, a variation H of σ induces the unique solution X of $KX + XK = q^{1/2}Hq^{1/2}$ and $\dot{A} = q^{-1/2}Xq^{-1/2}$. Differentiating $E = DA^2/\text{tr}(A^2)$ gives

$$\dot{E} = D \left(\frac{A\dot{A} + \dot{A}A}{\text{tr}(A^2)} - \frac{2A^2 \text{tr}(A\dot{A})}{\text{tr}(A^2)^2} \right).$$

Thus the smooth inverse and compact positive domain give the asserted nonsingular Jacobian bound. This calculation is finite-dimensional; it makes no claim about unbounded-operator closures.

12. BLIND TRANSPORT AND SINGULAR ACQUISITION

The exact criterion also applies when every displayed one-step stability coefficient is one, and when no affine density exists. This gives a sharp law different from the smoothing law of Section 10.

Let X have a specified prepared law μ on a compact subset E of a finite-dimensional Euclidean space. A protocol consists of one paid capture and $n - 1$ paid blind calls. Only the action *capture* is legal initially; it reports X exactly. At subsequent nonterminal phases only *advance* is legal and its report is a fixed symbol. The physical variable X remains unchanged. After exactly n calls the observer predicts X under squared loss. The terminal scored outcome records X after the decision, so bounded continuous functions of that outcome are executable terminal tests. It is unavailable before the prediction; an additional physical audit call, when needed, is counted separately. Thus the capture/advance interface is part of the experiment, not an uncounted clock supplied to the observer.

Define the actual-law quantization error

$$q_L(\mu) = \inf_{c_1, \dots, c_L \in \text{conv}(E)} \int \min_j \|x - c_j\|_2^2 \mu(dx), \quad L \geq 1.$$

The centers may be restricted to the compact convex hull, so a minimizer exists. The countable future tests on E give the compact predictive carrier $\mathcal{P}(E)$; the capture law on this carrier is the pushforward of μ by $x \mapsto \delta_x$. Blind updates are the identity. Affine mixture closure holds for both raw kernels. No ambient smooth parameterization of μ is used.

Theorem 12.1 (Exact blind-transport law). *For this experiment, every external width profile $\mathbf{m} = (m_1, \dots, m_n)$ satisfies*

$$(60) \quad R_{\text{ext}}(\mathbf{m}) = q_{\min_t m_t}(\mu).$$

The stationary internally exact- n class is nonempty precisely when $M \geq n + 1$, and in that range

$$(61) \quad R_{\text{aut}}(n, M) = q_{\lfloor (M-1)/n \rfloor}(\mu).$$

In particular, these conclusions hold for singular continuous and countably atomic acquired laws. They are assertions for the declared protocol and task, not for an experiment permitting repeated captures.

Proof. Fix a cut $t \geq 1$. All future reports are constant, and fresh transition randomness is independent of X conditional on the current label Z_t . Hence the terminal prediction U satisfies the Markov relation $X - Z_t - U$. Conditional squared-loss decomposition gives

$$\mathbb{E}\|X - U\|^2 \geq \mathbb{E}\|X - \mathbb{E}[X \mid Z_t]\|^2 \geq q_{|\text{supp } Z_t|}(\mu).$$

The last inequality remains valid for randomized encoders: the conditional means give at most $|\text{supp } Z_t|$ centers, and pointwise least squared distance is no larger than the encoder's conditional average. This lower holds at every cut. Conversely, quantize X once using $L = \min_t m_t$ centers and carry the resulting label unchanged through all blind calls. The external clock supplies the legal phase and terminal request. This proves (60).

For the autonomous machine, its initial label cannot occur later with positive probability. Its action must be capture initially and advance at every later nonterminal phase; the same stationary action kernel cannot do both. Nor can a nonterminal label be a stopping label. Between cuts 1 and n , the reports are identical and the label dynamics are stationary. If one positive-probability label occurred at two different such cuts, its law of future stopping time would be the same from both, contradicting exact stopping at n . Thus these n cut supports and the initial label are disjoint. Their sizes $L_t \geq 1$ satisfy $1 + \sum_{t=1}^n L_t \leq M$, so one is at most $\lfloor (M-1)/n \rfloor$. The cut lower just proved gives (61). To attain it, use one initial capture label and n disjoint copies of an optimal L -point quantizer label. The first copy is selected by the capture report, every blind transition advances its phase copy without changing its quantizer index, and the last copy stops and emits the center. The total number of states is $1 + nL \leq M$. The same construction with $L = 1$ proves nonemptiness at $n + 1$, and the disjoint-support argument proves necessity. $\square$

This theorem is also an explicit solution of the allocation frontier in Theorem 5.1. At capture, the law of conditional distributions is a barycentric contraction of the actual law of δ_X . At every later cut a further contraction can only lose information. Carrying the chosen conditional distribution without further pooling makes all later Jensen defects zero, although the blind update has no strict contraction at all. Thus nonsummable bounds on generic Lipschitz sensitivities do not preclude exact causal completion. Conversely, the narrowest retained cut gives a lower unrelated to an artificial terminal density.

Corollary 12.2 (Fractal and accumulating atomic laws). (i) *Let μ be the equal-weight two-map Cantor law with contraction $r \in (0, 1/2)$ and set $s = \log 2 / \log(1/r)$. Then*

$$q_L(\mu) \asymp_r L^{-2/s}.$$

(ii) *Let $\alpha > 0$, $x_j = 2^{-j}$, and*

$$\mu\{x_j\} = (1 - 2^{-\alpha})2^{-\alpha(j-1)}, \quad j \geq 1.$$

On the compact support $\{0\} \cup \{x_j : j \geq 1\}$ one has

$$q_L(\mu) \asymp_\alpha 2^{-(\alpha+2)L}.$$

Substitution into (60)–(61) gives matching causal laws, with the actual prepared masses unchanged.

Proof. For (i), a level- k cylinder has diameter r^k and mass 2^{-k} . With $k = \lfloor \log_2 L \rfloor$, its 2^k cylinder representatives give $q_L \leq r^{2k} \leq C_r L^{-2/s}$. The gaps separating neighboring cylinders at their first distinct level imply $\mu(B(x, \rho)) \leq C_r \rho^s$ for every x and $0 < \rho \leq 1$: choose k with $r^{k+1} < \rho \leq r^k$; an interval of length 2ρ meets only a bounded number, depending on r , of level- k cylinders. For L centers choose $\rho = (2C_r L)^{-1/s}$, after increasing the constant for the finitely many coarse radii. At least half the acquired mass is outside their balls. This gives $q_L \geq \rho^2/2$.

For (ii), retain the first $L - 1$ atoms exactly and place the remaining center at zero. Then

$$q_L \leq \sum_{j \geq L} (1 - 2^{-\alpha})2^{-\alpha(j-1)}2^{-2j} \leq C_\alpha 2^{-(\alpha+2)L}.$$

The $L + 1$ intervals centered at x_j with radii $x_j/8$, $1 \leq j \leq L + 1$, are pairwise disjoint. At least one contains no one of L proposed centers. Its atom alone contributes at least $c_\alpha 2^{-(\alpha+2)(L+1)}$ to the squared error. This lower is uniform over centers and proves the claim. $\square$

The support in (ii) accumulates, and its effective finite-resolution curve is exponential rather than the power predicted by an ambient one-dimensional chart. Its proof uses the actual atom masses. The result does not assert a corresponding formula for arbitrary singular laws: for those, Theorem 12.1 retains the exact quantity $q_L(\mu)$.

In Theorem 12.1, m_n counts terminal stopping/readout labels and the decision may be issued immediately upon entering one. The post-decision audit cannot break the cut Markov relation $X - Z_t - U$. In the accumulating-atom lower bound, absence of a center from the selected interval means that every center is at distance at least $x_j/8$ from its atom; this gives exactly the asserted single-atom contribution.

13. CAUSAL SIMULATION AND FINITE EXECUTION

The preceding cardinalities count persistent labels. A programme describing real-valued functions or arbitrarily long tables is not free bit complexity. We separate statistical comparison, numerical execution, and one finite class where synthesis is effective.

Proposition 13.1 (Typed common-task transfer). *Let $\mathfrak{E}$ and $\mathfrak{F}$ be prepared experiments with the same bounded scored task, $0 \leq L \leq H$. Suppose a parameter-independent causal simulator implements every legal target strategy in a specified class, including its action interface, reports, stopping rule, and task readout. Assume it uses at most R internal labels, transforms the target acquisition bound N to a declared physical bound $C(N)$, and gives total variation at most ε between the complete stopped and scored laws, uniformly in the compared preparations and strategies. Then*

$$(62) \quad \mathcal{R}_{\mathfrak{E}}(C(N), RM) \leq \mathcal{R}_{\mathfrak{F}}(N, M) + H\varepsilon.$$

Two such simulators compose with the product of their internal cardinalities, composition of their acquisition/physical-time maps, and the sum of their complete-law TV defects. A reverse risk lower requires a reverse certificate covering every strategy in the class to which the lower is applied.

Proof. For each target observer store the pair of its current label and the simulator label. The causal strategy lift supplies only legal actions, transforms reports at the promised completion times, and uses the same scored readout; the direct persistent cardinality is at most RM . A bounded loss changes by at most H times total variation, so taking infima proves (62). For composition, insert the intermediate simulated law, use contraction of total variation under the next parameter-independent kernel, and then its own defect bound. The product-state and composed-call accounts follow from the direct implementation. Without a reverse strategy lift this construction says nothing about source strategies not obtained from a target observer, which proves the stated converse qualification. $\square$

Programme descriptions concatenate, scratch workspaces add in a direct simultaneous implementation (or can be reused when their lifetimes are proved disjoint), and arithmetic and physical execution times add along the actual calls. An externally supplied stage index is not automatically supplied by a simulator of an internally timed task. The proposition compares total risks of a common task; subtracting two different full-history baselines would not preserve the inequality. Marginal report TV, an unscored transcript comparison, or a fixed-time simulator without stopping compatibility is insufficient. No large-deviation equivalence is inferred from this proposition.

Proposition 13.2 (Certified execution error). *Consider an ideal finite observer for a task bounded by H and at most n physical transitions. Suppose an implementation has a coupling with the ideal process such that, while the complete pasts agree, the next physical/label transition fails to agree with conditional probability at most e_t . A bound under the ideal prefix law on the corresponding first-mismatch event is also sufficient. If the final readout perturbation changes the conditional task loss by at most $\omega(v)$, then*

$$(63) \quad R_{\text{impl}} \leq R_{\text{ideal}} + H \min\left\{1, \sum_{t=0}^{n-1} e_t\right\} + \omega(v).$$

An ideal observer using approximate Bellman selections adds its actual expected action slacks $\sum_t A_t$ as in (31); these slacks need not be counted a second time in e_t .

Proof. Partition the event that the paths disagree by the first transition at which they do. Before that time both constructions have the ideal prefix, so the probability is at most $\sum_t e_t$, capped at one. On the complementary event the scored outcome is the same apart from the asserted readout change. The loss bound proves the inequality. The action-slack statement is the exact telescope, not a stability estimate between separately optimized controllers. $\square$

For the grids of Section 10, a coordinate error at most b_t can change a cell only in slabs of total ideal acquired mass at most $\min\{1, C_d \rho b_t / h_t\}$, for $b_t \leq h_t$, plus a separately certified exceptional evaluation probability. Coordinates mean the fixed affine simplex or traceless-matrix chart used for volume and grid cells. These are unconditional ideal masses; conditioning on survival of earlier numerical comparisons would bias them. Certified kernel couplings, integration/barycenter routines, Bellman selection errors, report precision, programme bits, scratch space and time must be supplied to apply this observation. It is not an algorithm for arbitrary Borel inputs. Predictive-coordinate error used to choose a cell is distinct from a Brier probability-readout error, whose loss modulus on a bounded probability simplex is linear in its Euclidean size.

Theorem 13.3 (Effective finite rational compilation). *Suppose the physical state, action, report and terminal-decision sets are finite, with cardinalities S, A, Y, U , respectively. Preparation, phase-indexed raw transition probabilities and bounded losses are rational, and n, M are fixed integers. For the stationary machine convention of Definition 2.1, including the phase-dependent physical legality constraints, nonemptiness and the optimum risk are decidable; whenever nonempty the*

optimum is attained by a machine with real algebraic row probabilities. No efficiency bound for the synthesis is asserted.

For every integer $Q \geq 1$, this machine admits a rational, support-preserving rounding with denominator Q , at the same M , which is still legal and stops exactly at n , with risk increase at most

$$(64) \quad H \min\{1, n(A + M)/Q\}.$$

It can be stored using

$$O((MA + M^2AY) \log(Q + 1) + M \log(U + 1) + M)$$

programme bits, in addition to the given model and a fixed interpreter. Its transition scratch space is $O(\log(Q + 1) + \log(M + 1) + \log(A + 1) + \log(Y + 1))$ bits. Exact rational sampling has finite expected runtime; limiting each sample to k rejection trials yields a bounded runtime and an additional risk error at most $2Hn2^{-k}$, without violating legality or the deadline. The program-size statement is a feasible upper, not an optimal description-length law.

Proof. Enumerate the finite choices of initial label, stopping subset and terminal decisions. For one such choice the unknowns are the stationary probabilities $\alpha_z(a)$ and $k(j \mid z, a, y)$, each in a finite simplex. Enumerating length-at-most- n physical/label paths expresses their probabilities as polynomials with rational coefficients. Requiring zero probability of every premature stop, illegal action and non-stop at n gives polynomial equalities. Each forbidden-path probability is a sum of nonnegative monomials. Expected loss is a polynomial on this closed subset of the compact product of simplexes. Thus a minimum exists when the set is nonempty. Quantifier elimination over real closed fields decides feasibility, gives its optimal algebraic value and an algebraic minimizing point; take the minimum over the finitely many enumerated choices[2]. This procedure may be extremely expensive.

For each probability row of length r , round to integer numerators totaling Q without introducing mass outside its original support. Taking floors and distributing the remaining units only within that support gives total variation at most r/Q . Coupling the action and label rows at each step gives mismatch probability at most $(A + M)/Q$. Crucially, support-preserving rounding cannot create any forbidden physical/label path: every such path has a zero raw factor or a zero original controller factor, and these zero factors remain zero. Hence legality and exact stopping hold with probability one for the rounded machine, not merely approximately. Proposition 13.2 proves (64).

Store one action row per label and one next-label row for each (z, a, y) , together with stopping flags and decisions. This gives the displayed bit bound. To sample a denominator- Q row, draw $\lceil \log_2 Q \rceil$ fresh bits and reject values outside $\{0, \dots, Q - 1\}$. Acceptance probability is greater than one half, and cumulative sums find the selected outcome by scanning the row. Only an index, a sum and the sampled integer are needed in scratch space. The expected number of trials is less than two; row search uses $O((A + M) \log(Q + 1))$ elementary bit operations per paired action/transition sample, in addition to the raw physical call. After k rejections use a fixed member of the same support. Each fallback has probability at most 2^{-k} , so two samples per call add at most $2n2^{-k}$ to the coupling defect. The bounded-trial variant additionally uses $O(\log(k + 1))$ scratch bits for an erased trial counter. The number of sampling bit operations is then bounded by $O(n(k + A + M) \log(Q + 1))$. The fallback never introduces a new support edge and therefore preserves the exact interface. A trial counter is scratch space erased before the next physical call, not a persistent random seed. $\square$

Arbitrary compact Borel instruments do not inherit finite rational decidability. Theorem 8.2 instead gives effective value approximation for its separately certified continuous-instrument class. Signs and zero tests of algebraic row probabilities are decidable, so the support used in the finite rational rounding can be identified. Only paths up to the fixed horizon are enumerated; no

infinite-horizon recurrence constraint is solved. Bounded finite-state-controller parameter synthesis predates this compilation result[23, 25]; the additional execution statement here preserves support, legality and deadline at the declared budget.

Corollary 13.4 (Attained acquisition, memory and error coordinates). *Take a regular engine of Theorem 10.1, or a blind engine of Theorem 12.1. Before its $n \geq 1$ calls, make one paid early call observing an independent fair bit through an erasure channel with erasure probability 2ε , $0 \leq \varepsilon \leq 1/2$. An estimate of this bit must be written immediately to a write-only scoring port. Also prepare an independent offset taking $\pm\delta$ with equal probability, $0 \leq \delta \leq 1$, never reported to the predictor. Its squared prediction error is scored together with the engine's loss and the early bit error. One final paid audit is performed by the scoring authority after the predictor has halted and fixed all decisions, without releasing information beforehand. Thus $N = n + 2$ counts physical acquisitions and the predictor halts after $N - 1$ calls; it does not make a further transition for the audit.*

The source-to-perfect-bit causal deficiency at the early completion deadline is exactly ε , uniformly over the two bit values. For the autonomous class, all $M \geq N$ satisfy, in the regular case,

$$(65) \quad \mathcal{R}(N, M, \delta, \varepsilon) - B_{N-2}^* \asymp (M - N + 1)^{-2/d} + \delta^2 + \varepsilon,$$

and, in the blind case, the exact identity is

$$(66) \quad \mathcal{R}(N, M, \delta, \varepsilon) = q_{\lfloor (M-2)/(N-2) \rfloor}(\mu) + \delta^2 + \varepsilon/2.$$

The constants in (65) are those of the fixed regular class, independent of these budgets and errors. The calibration and deficiency terms are actual inaccessible alternatives and actual interface defects, not arbitrary permitted error tolerances.

Proof. The best bit prediction on erasure is $1/2$, giving squared loss $\varepsilon/2$; the offset has conditional mean zero and variance δ^2 . These variables are independent of the engine and scored additively under one product law, so their lower bounds add to its risk for any observer. The physical interface permits only the early action at the initial label and only engine actions afterwards. That initial label is therefore distinct from all engine labels. The write-only output is not readable and cannot function as a persistent register. Any extra bit-dependent randomization of the engine can be generated as fresh independent initial randomization; it supplies no information about the independent engine variable and cannot improve the minimum over initial engine programs. Conditional on that seed, the corresponding legal engine begins in one of at most $M - 1$ labels. Taking the best such conditional program proves the engine lower with $M - 1$. The matching upper uses one early label and the corresponding engine machine with $M - 1$ labels; the bit estimate is emitted during the early transition. After the predictor halts and fixes its decisions, the scoring authority performs the final physical audit; this adds one charged acquisition but no predictor transition. Thus N counts physical acquisitions, while the autonomous predictor itself halts after $N - 1$ calls. This distinction is part of the stated task, not a free terminal signal during the engine. Applying the two engine laws gives the displayed formulas.

For the deficiency, the two erased-bit source report laws have TV distance $1 - 2\varepsilon$, whereas the perfect-bit laws have distance one. Any common parameter-independent postprocessing preserves the former upper bound. If its error from each perfect law were at most e , the triangle inequality would give $1 \leq 2e + 1 - 2\varepsilon$, hence $e \geq \varepsilon$. Reporting the observed bit and guessing fairly on erasure attains error ε for both values. The imposed early deadline forbids using the later audit. Physical preparation, erasures, engine calls and the final scoring call are all counted; the calibration offset is not an unknown kernel that may be learned from hidden free samples. $\square$

14. RELATION TO EARLIER THEORIES AND SCOPE

The static barycentric step belongs to the theory of comparison of experiments and convex order. Blackwell’s comparison theorem[3] concerns the decision content of a parameter-independent garbling; Strassen’s theorem[17] identifies convex order with a martingale coupling. Lemma 6.1 is a finite-target proof of this classical mechanism. Bayesian persuasion[8] optimizes information structures subject to Bayes plausibility; coarse information design[13] relates conditional-mean cells to the curvature of their value function. Here the incoming law is fixed by the actually executed physical acquisition. One may only garble that law, and the next law depends on the retained information and chosen action. Neither Bayes plausibility alone nor a static posterior partition enforces these serial constraints.

Conditional convex order and measurable kernel selection have an established theory. Leskelä and Vihola[24] prove conditional martingale-coupling results, including a kernel formulation. Our finite-target allocation proof does not claim measurable martingale selection as a new result. The serial restriction is that each next acquired law must be generated by the already retained law through a specified physical instrument. Sequential persuasion[22] studies serial signals with strategic obedience and commitment, whereas our external allocation has no separate strategic receiver: it is a nonstrategic finite-message allocation with the physical acquisition instrument fixed. A stationary signalling restriction is analogous to programme reuse, but no equivalence of the strategic and nonstrategic optimization problems is asserted. Dynamic information acquisition[27] optimizes sequences of information choices under information costs; in that theory the admissible signal design can be much broader than garbling a fixed instrument. Here a hard label cap and a common reused programme impose different constraints. Neither comparison identifies Bayes plausibility with executable causal allocation.

Causal transport[1] imposes nonanticipation on couplings and has its own dynamic programming principle. The present restriction is different from nonanticipation alone: retained information must have prescribed finite cardinalities, and a reused internal label must invoke one and the same report kernel at every phase. Equation (27) imposes this latter constraint simultaneously, rather than selecting an arbitrary adapted coupling at each time. The theorem does not identify directed information, a channel rate, or a transport cost with persistent label cardinality. The compact finite-obstruction result is a topological separation statement; it is not an efficient optimization algorithm over causal transport plans.

The most direct computational neighbor is bounded finite-state-controller synthesis. Poupart and Boutilier[25] optimize restricted stochastic controllers of POMDPs. Junges et al.[23] relate bounded randomized controllers to parameter synthesis for parametric Markov chains. Andriushchenko et al.[20] use oracle-guided inductive synthesis over finite-controller design spaces, including indefinite-horizon specifications; our exhaustive finite bound does not claim their algorithmic efficiency or scope. Amato, Bonet and Zilberstein[21] compare Mealy and Moore controller conventions, so the timing of report access and action output is part of the resource model, not innocuous notation. Our convention is a state-based action row followed by a report-dependent update row: the report dependence is Mealy-like in the update, whereas the action is tied to the current location. A different Moore/Mealy timing convention can change the charged number of locations. We do not claim that bounded-controller decidability or finite polynomial path constraints are new. Equation (36) exhibits the fixed-proposal linear system, and Theorem 13.3 is positioned as a finite rational execution specialization. The general simultaneous theorem eliminates the same observation-dependent row across all occupied phases on compact predictive carriers. It does not remove the nonlinear optimization over the proposed conditional occupancies.

Robust finite memory is not a new subject. Cubuktepe et al.[28] formulate robust finite-state-controller synthesis for uncertain finite POMDPs, dualize a semi-infinite nonconvex problem,

and obtain locally optimized controllers by sequential convex programming. Their model also uses support-preserving uncertainty assumptions. Our comparison is not “uncertainty versus known models”: both theories constrain one controller across model uncertainty. The additional statement here is a fixed-horizon compact positive-instrument theorem with continuous report spaces, an exact common-parameter/common-phase moment elimination, global finite-model obstructions, and a supplied uniform error bound relative to all Borel controllers at the same state budget. Its exhaustive search makes no claim of matching the scalability of their synthesis method. Their finite reward/reachability specifications and controller timing conventions must not be silently identified with our exact-deadline terminal task.

Hudák et al.[29] train neural policies, extract finite-state controllers, and evaluate them formally; their hidden-model extension seeks robust controllers across finite sets of POMDPs. Thus neither extracting a robust controller nor validating its performance is claimed here as new. Theorem 3.3 permits jointly dominated, possibly uncountable model sets and gives finite exclusion of every programme below a strict target. Theorem 3.1 then makes the entire fixed-budget optimum effectively approximable under uniform raw-instrument moduli. These are mathematical global-comparison statements, not competing empirical scalability claims. The unknown parameter is fixed for the entire experiment; we do not identify it with a dynamically changing adversary.

Several approximation theories quantify finite-memory performance for known POMDPs. Kara and Yüksel[9] construct finite-window approximations under filter stability and finite observation/action alphabets. Demirci, Kara and Yüksel[5] refine such estimates using expected Wasserstein or pathwise total-variation stability. Saldi, Yüksel and Linder[15] establish finite-model approximation for discounted partially observed control under continuity conditions. These approaches compare approximate policies or values with their control benchmarks. They do not impose the same internally counted exact-deadline interface used in (52). Our quantitative theorem instead combines an actual-law converse with explicitly allocated finite observer states, while the exact allocation theorem itself does not require their stability conditions. This distinction does not imply that the present theorem solves the discounted or average-cost problems treated in those works.

Observation quantization itself is likewise not new. Kara, Bayraktar and Yüksel[26] discretize continuous observations and use finite histories under controlled filter-stability conditions. Their policy near-optimality and learning results concern a different control benchmark. Theorem 8.2 preserves a fixed whole-controller alphabet and supplies lower as well as upper certificates relative to all Borel controllers at that budget. It needs a finite horizon, effective raw integrals, a common reconstruction and support data, but not filter contraction. Its exhaustive finite search is not a more efficient planning algorithm, and it does not subsume discounted or reinforcement-learning conclusions. Positive-instrument comparison, probability-row rounding and finite enumeration are standard ingredients; the proposed new assertion is their uniform, same-budget, exact-interface combination.

Grover and Dimitrakakis[7] adapt belief discretization to depth in POMDP planning. Li, Hammar and Bertsekas[12] aggregate beliefs through features and conditional means and derive value-approximation bounds. Adaptive grid size, feature aggregation and conditional means are therefore not new ingredients here. The additional requirement is calibration under all incoming edges of the deployed observer, and for an internally timed observer the compatibility of the same transition programme across all cuts. The mass-curvature estimate and block allocation used in Section 10 are inherited quantitative mechanisms[14]; they are not counted again as new results merely because they now follow an exact feasibility theorem.

The recent belief-metric covering analysis of Zhu and Lu[19] addresses off-policy evaluation and coverage/sample-efficiency bounds in offline POMDP learning. Its geometric coverage condition is not the hard cardinality of a causal observer’s persistent state. Kim[10] studies finite-memory

belief mismatch and policy-conditional performance along a common closed-loop execution. That information pattern differs both from optimizing over every legal controller and from charging a stationary machine’s timing states. These comparisons are at the stated theorem objectives; neither paper is dismissed on the basis of its terminology or publication date. In particular, this article does not claim matching acquisition complexity for learning an unknown raw kernel from offline data.

Zero-delay source coding is another close but distinct theory. Wood, Linder and Yüksel[18] and Ghomi, Linder and Yüksel[6] study optimal stationary codes and finite-memory approximations for Markov sources under long-run distortion criteria. Stavrou, Charalambous, Charalambous and Loyka[16] develop nonanticipative rate-distortion bounds and causal filtering constructions, including Gaussian processes. An information-rate or communication-alphabet constraint does not bound every encoder/decoder internal state. Conversely, our observer need not transmit one symbol over a communication channel at every physical call. Thus their infinite-horizon and information-rate statements cannot be substituted for the finite-label cut converse here, and our results do not subsume theirs.

Positive quantum instruments have a standard operational basis[4]. Theorem 11.3 uses a deliberately explicit finite-dimensional positive preparation and spanning instrument, so that quotient separation, report continuity and acquired density can be verified without commutativity. It is not a theorem about arbitrary quantum memory, operator domains, or physically unrestricted measurement design. Likewise, Proposition 13.1 is the usual bounded-risk comparison mechanism[11] with the missing causal and resource interfaces made explicit, rather than a new total-variation inequality. The effective finite compilation theorem uses classical real algebraic decision methods[2]; it gives no new complexity bound for quantifier elimination.

The present principal addition is the shared-parameter extension of the causal frontier, its positive-instrument attainment and finite-model obstruction, and its uniformly effective minimax completion. The singleton effective theorem, with common report reconstruction and support-preserving rounding, is retained and re-proved as a constituent rather than counted again as new. The exact finite-support allocation frontier, its phase-compatible dual, compact external obstruction and distinct matching regimes are retained and re-proved. The zero-loss criterion is independent of full task curvature; the block-stable law and blind-transport law exhibit when, and why, quantitative conclusions differ. Full predictive quotients of arbitrary Borel experiments, a uniformly effective theory of arbitrary stationary compact-carrier optimizers without computation and support certificates, unknown-kernel minimax learning, and a matched optimal region for programme size, scratch space and physical time are not established. Neither the fixed-horizon theory nor its bounded-stopping error estimate is an infinite-horizon theorem.

The strengthened family criterion still has boundaries. Common domination makes one Borel representative valid for every model. The finite-model compactness proof uses a fixed finite-dimensional physical instrument representation, not compactness of arbitrary products of normalized filters. Effective global convergence further requires a common support table, computable model nets, uniform instrument moduli, and executable report cells. It need not hold for a merely pointwise computable family or preserve exact stopping under a small but support-changing approximation. These hypotheses are verified for the stated unknown Gaussian and quantum families; neither verification supplies a universal learning rate or a closed formula in parameter dimension. Priority for the precise combination remains subject to independent literature and mathematical review.

APPENDIX A. FROM RAW EXPERIMENTS TO A MEASURABLE PREDICTIVE QUOTIENT

We give a sufficient realization result at the raw level. It does not assert that every standard Borel predictive equivalence relation has a compact or even standard Borel quotient.

For each phase let X_t be compact metrizable. Let $K_t^a(x, dx' dy)$ be a positive normalized Borel physical kernel with standard Borel reports. Fix on each report space a compatible Polish topology and a countable bounded continuous measure-determining family containing constants and closed under rational linear combinations and products. Let $\mathcal{V}_t \subset C(X_t)$ be a separable closed linear subspace containing constants and the conditional expectations of a determining family of executable future tests. Assume it contains every pullback

$$(67) \quad x \mapsto \int \varphi(y) f(x') K_t^a(x, dx' dy), \quad f \in \mathcal{V}_{t+1},$$

for legal a and each function φ in the chosen report family. In particular these pullbacks are continuous. Assume the declared future-test system is determined by these tests and their successive report-contingent continuations; equality on them must not discard a legal action or an observable resource coordinate. Phase-dependent legality is retained separately. The terminal task has continuous affine expectation in the resulting coordinates. These are verifiable closure conditions on the raw experiment, not assumptions of a Fisher metric or an available exact posterior register.

Choose a countable norm-dense family $f_{t,i}$ in the unit ball of $\mathcal{V}_t$, including constants, and define

$$(68) \quad Q_t : \mathcal{P}(X_t) \longrightarrow [-1, 1]^{\mathbb{N}}, \quad Q_t(p) = (\int f_{t,i} dp)_{i \geq 1}, \quad K_t = Q_t(\mathcal{P}(X_t)).$$

The dense family may be chosen from the rational span of determining executable tests; equality of its coordinates is therefore equality of their predictions, not an extra observable added to the experiment.

Theorem A.1 (Predictive realization and minimality). *Under the preceding raw hypotheses, K_t is compact metrizable and convex. Two physical distributions have the same Q_t precisely when their declared executable future tests agree. The actual posterior of every legal finite-history strategy projects to a Borel K_t -valued predictive state. Report and update kernels on K_t admit jointly Borel versions, independent of the representing distribution up to the relevant report-null sets, and satisfy (2). Conditional mixing is realized by barycenters.*

Every measurable statistic determining all these future tests factors the projected predictive state through a Borel map; that factor is unique on the statistic's attained values. Conversely, the projected state determines the tests. This is minimality for the fixed preparation/test interface, not simultaneous frequentist sufficiency for all inaccessible parameters. If the induced acquired-law kernels are weakly continuous, all the hypotheses concerning predictive realization in Theorem 5.1 hold.

Proof. Weak compactness of $\mathcal{P}(X_t)$ and continuity of the coordinates make K_t compact in the countable product; it is convex by linearity. Equality of dense coordinates gives equality on $\mathcal{V}_t$. The report pullbacks with $f = 1$ determine the report law. With general f they determine the finite signed measures $B \mapsto \int_{X_t \times X_{t+1} \times B} f(x') p(dx) K_t^a(x, dx' dy)$. The measure-determining report family and a monotone-class argument extend the identities to all Borel report sets. After disintegration, equal incoming coordinates therefore give equal outgoing conditional coordinates outside one report-null set, chosen common for the countable family. Induction over the finite continuation proves equality of the declared future tests, including report-contingent choices. The reverse implication holds because the determining tests are executable. Continuity of arbitrary report-contingent test expectations is not needed for this implication.

For measurability, a continuous map from a compact metric space onto a compact metric image has a Borel section here. An explicit construction is useful. Cover $\mathcal{P}(X_t)$ at level m by finitely many closed balls of radius 2^{-m} . Select the least ball whose intersection with the previously selected compact intersections still meets the fiber $Q_t^{-1}(q)$. The condition is Borel in q , since the

image of each such compact intersection is closed. The nested nonempty compact intersections have diameters tending to zero, and their unique limiting point is a Borel selection $p(q)$ in the fiber.

Apply a jointly measurable conditional-probability construction to the raw kernel with input $(p(q), a)$. Such a construction follows, for example, by conditioning on successive finite partitions generating the report sigma field: conditional expectations of a countable dense family in $C(X_{t+1})$ converge by the martingale theorem, determine a probability measure on the compact state space almost everywhere, and have measurable limits. Assign an arbitrary fixed probability where the convergence or consistency tests fail. Project this posterior by Q_{t+1} to define $F_t(q, a, y)$, and use the raw report marginal to define $J_t(q, a)$. The preceding signed-measure identities prove representative independence almost everywhere. The initial and all later history spaces are standard Borel, so this construction also gives Borel projected posteriors along actual histories.

For a probability ν on K_t , mix its selected representatives: $p = \int p(q)\nu(dq)$. Then $Q_t(p) = \bar{\nu}$. Linearity of the unnormalized raw kernel in p , followed by the signed-measure identities, gives both formulas (2). Equality on coordinate functions is enough: it identifies each conditional barycenter in K_{t+1} and hence every continuous affine function. This is mathematical realizability as a conditional physical mixture. It does not make re-preparation free or assert that every point of the compact carrier is reached under the chosen exploration. In a feasible allocation flow the induction of Lemma 6.2 supplies the actual conditional mixtures under the deployed machine.

Finally, let a statistic $T(h)$ determine the tests, with Borel coordinate maps b_i such that $Q_t(p_h)_i = b_i(T(h))$. The vector $b = (b_i)$ is Borel. Since K_t is closed in the product, replace b by a fixed point outside the Borel set $b^{-1}(K_t)$. This defines a Borel factor through T agreeing with the projected state on every attained history where the conditional versions are specified. Its values there are forced by the separating coordinates. No claim that the image of an arbitrary history map is Borel is required. The remaining weak-continuity condition is additional; it is not a consequence of measurability alone. $\square$

In the finite hidden-state model, the test span identifies the simplex of distributions. In the noncommuting instrument model, all future-test expectations are affine in the density matrix and the spanning effects identify that matrix; different classical decompositions of it are correctly identified. For capture of a compactly supported variable, bounded continuous report tests identify its probability law, and the carrier can be infinite-dimensional. These identifications explain why the same abstract theorem covers the three settings without assigning an ambient Fisher or covariance geometry at the outset.

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